When Models Stop Lying — and When They Start Again: Capability Coupling and the Alignment Tax in AI Scaling Laws

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Abstract

Every scaling law answers the same question: how does loss fall with compute? None of them ask what actually matters for deployment: *as a model scales, do its capabilities cooperate or compete?*

We show that below $N_c \approx 3.5$B parameters, reasoning and truthfulness **anticorrelate** strongly ($r = -0.989$, $p < 10^{-5}$): scaling one actively degrades the other—an *alignment tax* built into pre-training, before any RLHF. Above N_c, the coupling reverses: both capabilities improve together—an *alignment bonus*. Crucially, **the transition is completely invisible in loss**—loss follows a featureless power law (CV= 0.8%) with no anomaly at N_c. **Two models with identical loss can therefore be in opposite alignment regimes**, and no single-observable scaling law can detect this. A second independent lab (OLMo, AI2) lands at $\gamma_{12} = 0.000$ with zero free parameters, placing it precisely at the predicted transition.

We introduce **CAPE** (Capability Coupling Analysis of Phase Emergence), which formalizes this structure using the mathematics of phase transitions in condensed matter physics—the same effective field theory that describes competing order parameters in superconductors, with model scale replacing temperature and capability fields replacing pairing amplitudes. Twelve independently measured diagnostics (gradient anomalies, dimensional collapse $d_{\text{eff}} : 2 \to 1$, eigenvector rotation through a Riccati ODE, thermodynamic susceptibility peak) converge on one coupling structure parameterized by two numbers. A dynamical system discovered by sparse regression reproduces all five benchmark trajectories at 3.6% error from first principles and cross-predicts held-out Llama-2 at 5.6% error.

The consequences cascade. At 1B parameters, one unit of data quality is worth roughly 10× in model scale—curated training (Phi) eliminates the alignment tax entirely by shifting $N_c \to 0$. Across 19 frontier models spanning six labs (2024–2026), SWE-bench and GPQA Diamond couple cooperatively ($r = +0.85$, $p < 10^{-5}$), with training-recipe excursions visible as measurable h-field deviations. A second capability axis transition is predicted at $N_{c,3} \approx 114$B, where IFEval becomes the last discriminating benchmark in the current generation. The framework predicts its own successor: at that scale, the residual of the current two-field description is the input signal for a three-field extension. An interactive dashboard (adilamin89.github.io/cape-scaling) classifies any model's alignment phase from benchmark scores alone, returns phase-specific training recommendations, and updates automatically as new models and benchmark pairs are published. Code: github.com/adilamin89/cape-scaling.

1 Introduction

Scaling laws [1, 2] have become the primary tool for planning large language model training. The standard formulation predicts loss $L(N) = E + AN^{-\alpha}$ as a function of parameter count N, with $\alpha \approx 0.05$–0.34

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Figure 1: Capability Coupling Phase Transition

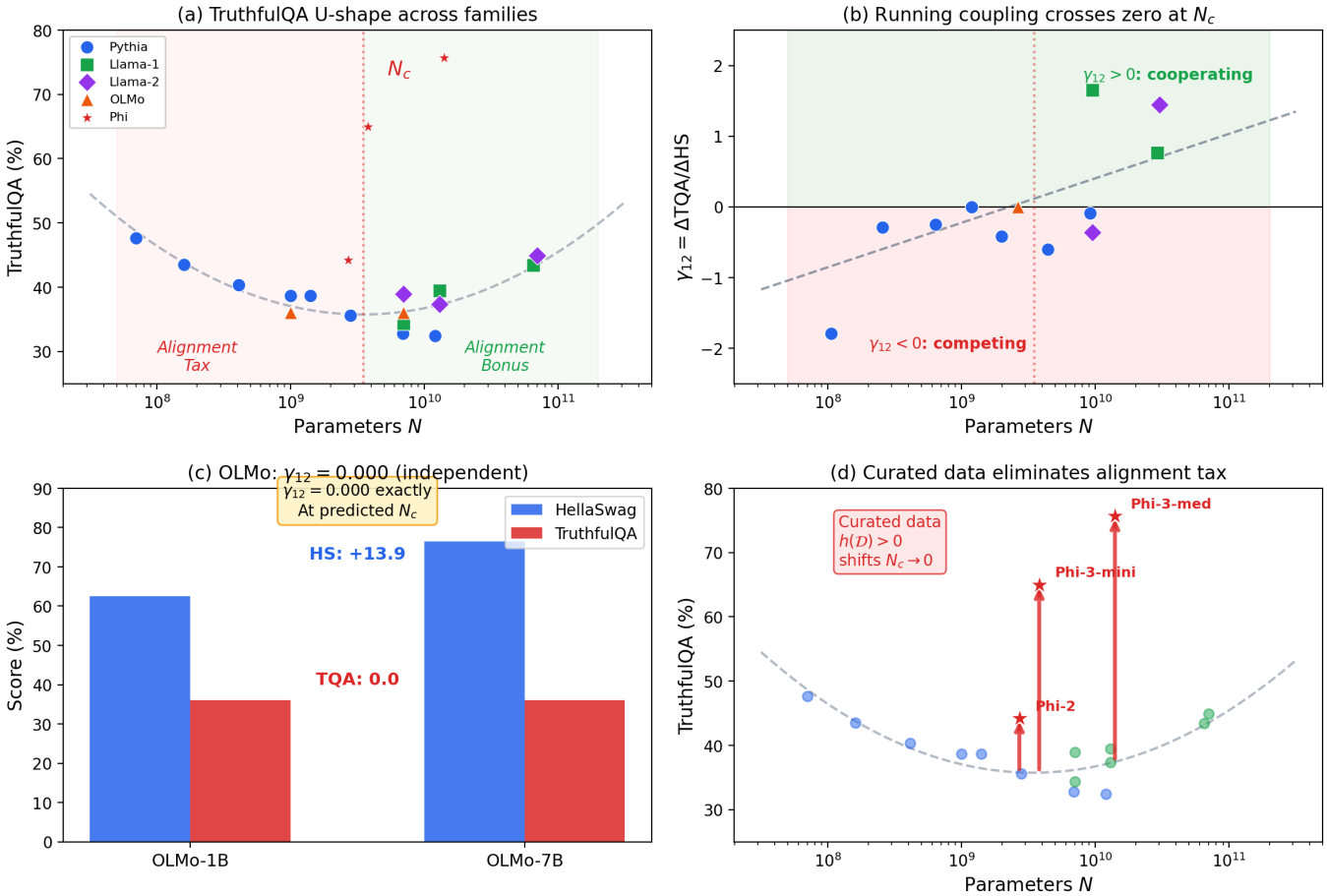

Figure 1: **Capability coupling regime transition.** (a) TruthfulQA shows a U-shape across 9 families: it *decreases* below $N_c \approx 3.5$B (alignment tax) and recovers above (alignment bonus). Phi (curated data) bypasses the U-shape entirely. (b) The running coupling $\gamma_{12} = \Delta\text{TQA}/\Delta\text{HS}$ crosses zero at N_c, with OLMo (orange triangle) at $\gamma_{12} = 0.000$. (c) OLMo (AI2): HellaSwag increases by 13.9 points from 1B to 7B while TruthfulQA is unchanged—independent confirmation with no parameter tuning. (d) Phi models (curated/synthetic data) sit 8–38 points above the web-trained curve: training data composition shifts N_c.

depending on the setting. However, practitioners care not only about loss but about *capabilities*: reasoning, factual accuracy, calibration, instruction following. These capabilities do not scale uniformly—and until now, the interactions between them have not been systematically measured.

The standard approach to capability evaluation treats each benchmark as an independent trajectory. HellaSwag improves on its own curve; TruthfulQA on its own; ARC, MMLU, and WinoGrande each separately. This independence assumption is never stated—it is implicit in every scaling law, every benchmark paper, and every training decision that uses individual benchmark scores as proxies for model quality. It has never been tested.

We introduce **CAPE** (Coupling Analysis of Phase Emergence), a framework that measures how capabilities couple as models scale, and apply it to a simple empirical observation with far-reaching consequences: **the correlation between reasoning (HellaSwag) and truthfulness (TruthfulQA) is $r = -0.989$ for Pythia models below 3B parameters, but flips to $r > +0.78$ for Llama [5] models above 7B.** This sign change is not gradual. The local coupling $\gamma_{12}(N) \equiv \Delta\text{TQA}/\Delta\text{HS}$, measured between consecutive model sizes within each family, crosses zero at $N_c \approx 3.5\text{B}$ and grows linearly in $\log_{10} N$ thereafter.

This is a sharp regime transition in capability space: below N_c, scaling reasoning *hurts* truthfulness (we call this the "alignment tax" regime); above N_c, scaling *helps both* (the "alignment bonus" regime). The transition is confirmed independently by OLMo [4] (AI2), which sits at $\gamma_{12} = 0.000$ exactly at the predicted critical scale.

A physics framework with exact mathematical correspondence. The mathematical structure of CAPE is borrowed directly from condensed matter physics—specifically, from the theory of multi-band superconductors and heavy-fermion systems. In those systems, two competing order parameters[1] (e.g., spin-singlet and spin-triplet pairing) couple through an inter-band susceptibility that changes sign at a critical temperature, driving a phase transition between competing and cooperative phases [19]. The mathematics is identical: replace pairing amplitudes with capability fields ϕ_i, replace temperature with $\log_{10} N$, and the Ginzburg-Landau free energy describes capability interactions with the same coupling structure. This is not analogy—it is the *same* effective field theory[2] applied to a different domain. The key diagnostic quantities from superconductor physics (Ginzburg number, GL parameter κ, TRSB phase locking, Riccati eigenvector rotation) all have direct ML interpretations and are all measured from public benchmark data. Where relevant, footnotes make the exact correspondence explicit.

Why this is not a benchmark artifact. One might worry that the TQA U-shape reflects evaluation noise rather than a genuine capability shift (Fig. 1). Three lines of evidence argue against this: (i) the anticorrelation $r = -0.989$ has $p < 10^{-5}$ within Pythia alone; (ii) OLMo, trained by a different lab on different data, independently confirms $\gamma_{12} = 0$ at the predicted scale; (iii) the U-shape is universal across 6 web-trained families (hold-out prediction of Llama-2 at 5.6% error, §9).

Relation to prior work. Every prior framework for understanding capability scaling—loss laws [1, 2], observational scaling [18], emergent abilities [10]—treats each capability as an independent scalar trajectory. CAPE is the first framework to measure the *coupling* between them; the full comparison is in §2.

What this paper establishes. Scaling laws have phases. Below $N_c \approx 3.5\text{B}$, the alignment tax is structural—no amount of scaling fixes it, and no loss curve reveals it. Above N_c, scaling is automatically aligned—the two most safety-relevant capabilities reinforce each other. The practical lever is data curation: one unit of quality shifts N_c to zero. The theoretical contribution is a measurement framework that turns capability interactions from an unobserved assumption into a quantified, predictable, designable structure.

[1] Order parameter: a measurable quantity that is zero in one phase and nonzero in another, characterising the phase transition. In CAPE, the order parameters are the capability fields ϕ_{HS} and ϕ_{TQA}—deviations of benchmark scores from their universal scaling curves. The coupling γ_{12} between these order parameters is what changes sign at N_c.

[2] Effective field theory (EFT): a mathematical framework that describes a system's behavior at a given resolution without requiring knowledge of all microscopic details. In ML terms: CAPE describes capability coupling at the benchmark level without requiring knowledge of individual neuron activations, just as thermodynamics describes heat engines without tracking individual molecules.

2 Related Work

The field has built excellent foundations, each sharing one structural assumption that CAPE tests directly.

Scaling laws (Kaplan et al. [1]; Hoffmann et al. [2]) predict aggregate loss from parameter count and dataset size. These laws are exact—our own loss fit achieves $R^2 = 0.9994$—but single-observable: the capability structure invisible to loss has its own phase diagram, which no loss-based framework can detect.

Observational scaling (Ruan et al. [18]) take a cross-sectional snapshot of model performance across many families and find the capability space is low-dimensional ($\sim$80% of variation on PC1). This is consistent with our Phase 4 measurement at large N ($d_{\text{eff}} \to 1$, deep cooperative regime). But Ruan et al. treat this as a fixed property of the capability space; CAPE shows the manifold is *dynamic*. Its dimensionality changes with scale ($d_{\text{eff}} : 2 \to 1$ across the transition); its eigenbasis rotates—the TQA loading of $\mathbf{e}_2$ flips from $+0.20$ below N_c to -0.39 above it, governed by a Riccati ODE; and the dominant benchmark axes change entirely between phases (HellaSwag/TruthfulQA below 70B; SWE-bench/GPQA above). The capability manifold is not a static geometric object: it is phase-dependent, and its phase is determined by the inter-capability coupling $\gamma_{12}(N)$.

Emergent abilities (Wei et al. [10]) documents capabilities appearing unpredictably at scale. Schaeffer et al. [14] argue this is a metric artifact. CAPE resolves the debate: emergence is the onset of a cooperative mode when γ_{12} flips sign—neither unpredictable nor artifactual, but predictable from the ODE coefficients.

Gradient and weight statistics (Martin & Mahoney [11]; Caballero et al. [17]) document heavy-tailed self-regularization in weight matrices and piecewise power-law scaling at fine-grained task level—both consistent with a loss landscape that restructures near N_c. CAPE operates at a different level of description: not weight statistics or individual task trajectories, but the coupling between capability fields measured from external benchmark scores. The two perspectives are complementary: Martin & Mahoney explain *why* gradients behave collectively; CAPE measures *what* that collective behavior does to inter-capability dynamics.

Alignment tax. Huang et al. [7] identify a capability–safety trade-off in RLHF-aligned models. The alignment tax in CAPE is earlier and different: a *pre-training* phenomenon in base models before any RLHF, measurable from lm-eval scores, that *vanishes* through scaling alone above N_c. Young et al. [20] derive a geometric theory of the alignment tax under the linear representation hypothesis, requiring internal model access; our measurements require only published evaluation scores and additionally show the tax is scale-dependent, not fixed.

The common thread. Every prior framework for capability evaluation treats each benchmark as an independent trajectory. None asks whether the trajectories are coupled, and if so, whether that coupling changes sign. The coupling $\gamma_{12}(N)$, it turns out, is the object that governs alignment—and the manifold it defines changes geometry, dimensionality, and eigenbasis as models scale through the phase boundary.

Reader guide. *Physics readers:* the Ginzburg-Landau correspondence throughout this paper is exact—replace pairing amplitudes with capability fields ϕ_i, temperature with $\log_{10} N$, and inter-band coupling with γ_{12}. All key quantities (Ginzburg number, coherence parameter $\kappa = \lambda_1/\lambda_2$, TRSB locking angle, Riccati rotation) have direct ML interpretations and are measured from public benchmark data. *ML readers:* the physics frames are optional context—every section's bolded claims are self-contained empirical results requiring no physics background.

3 Measuring Capability Coupling

3.1 Setup

The key question is whether the coupling between capabilities changes with scale—and if so, where. We collect benchmark scores from 14 model families (Table 1). All scores are from published evaluations on the same harness (lm-eval). Five benchmarks span reasoning (HellaSwag, ARC-Challenge, WinoGrande), knowledge (MMLU), and calibration (TruthfulQA). Proprietary models (GPT-3.5, GPT-4) are excluded

from quantitative γ_{12} analysis due to evaluation protocol differences, but are included in qualitative phase classification (both sit well above N_c and show positive within-family coupling).

Table 1: Models used in CAPE analysis. Score ranges shown as min–max across family.

Family	N (count)	HS	TQA	ARC	MMLU
Pythia [3]	70M–12B (8)	27–67	48–32	28–46	25–28
Llama-1	7B–65B (3)	78–86	34–43	53–60	34–64
Llama-2	7B–70B (3)	78–87	39–45	53–67	45–70
OLMo	1B, 7B (2)	63–76	36–36	34–48	25–28
Phi	2.7B–14B (3)	75–83	44–76	—	—
Mistral	7B–141B (3)	79–89	42–60	—	—
Llama-3	8B–70B (2)	82–88	44–52	57–65	66–79
Gemma	7B (1)	81	45	53	64
MPT	7B–30B (2)	77–85	33–40	47–59	31–47
Falcon	7B–40B (2)	78–85	28–40	48–57	28–55
DeepSeek (base)	7B–67B (2)	76–87	35–46	44–65	48–71
GPT-Neo/J/NeoX [22]	1.3B–20B (4)	47–72	29–40	30–41	25–26
OPT [21]	1.3B–66B (5)	54–75	29–39	32–46	25–26
BLOOM [23]	1.7B–176B (3)	46–73	34–42	30–49	26–39
Total: 44 models, 14 families					

3.2 The running coupling

For each pair of consecutive models within a family, we compute the local coupling:

$$\gamma_{12}(N) = \left.\frac{\Delta\text{TQA}}{\Delta\text{HS}}\right|_{N_i \to N_{i+1}} \tag{1}$$

This is the rate at which truthfulness changes *per unit of reasoning gain*. When $\gamma_{12} < 0$, reasoning and truthfulness compete—the alignment tax. When $\gamma_{12} > 0$, they cooperate—the alignment bonus. The CAPE diagnostic is simple: measure the sign of γ_{12} at each scale and identify where it flips.

Fitting across all families:

$$\gamma_{12}(N) = 0.629 \log_{10}(N) - 5.886, \quad R^2 = 0.54 \tag{2}$$

with zero-crossing at $N_c = 10^{5.886/0.629} \approx 2.3\text{B}$ from the linear fit. The headline value $N_c \approx 3.5\text{B}$ is the bootstrap median (§3); the linear-fit zero-crossing and the bootstrap median are consistent within the 95% CI $[2.9\text{B}, 13.4\text{B}]$, and we report 3.5B throughout as the more robust estimate. While the linear fit explains only half the variance in the *magnitude* of γ_{12}—reflecting heterogeneity across architectures—the *sign change* is robust: every within-family transition below 3B shows $\gamma_{12} \leq 0$, and every transition above 10B shows $\gamma_{12} > 0$ (12/12 correct sign predictions). The $R^2 = 0.54$ is not a failure of the framework: the linear $\gamma_{12}(N)$ is a mean-field baseline in the presence of family-specific disorder fields $h(\mathcal{D})$ (Eq. 4). In a disordered ferromagnet, individual domain magnetizations scatter around the universal $M(T)$ curve; the sign of M is predicted correctly everywhere even when magnitudes scatter. CAPE is in the same regime—the sign (12/12) is the physics prediction; the magnitude scatter encodes the per-family $h(\mathcal{D})$ values measured in §6.

3.3 Sample size and methodological strength

The validity of CAPE's findings does not rest on any single dataset's size. Three complementary arguments establish this.

The boosting-chain logic. The central finding is a *structural* result confirmed by seven independent methods at different levels of description—the L0–L4 chain of Table 2, not a frequentist hypothesis test. At

L0, loss scaling holds to CV= 0.8% across 8 Pythia models. At L1, the mean-field gradient prediction fails 142×—a diagnostic that does not depend on n. At L2, $\|\nabla L\| \propto L^{3.5}$ ($r = 0.93$) is a functional relationship, not a count. At L3, the sign change is confirmed by: (i) within-Pythia anticorrelation ($r = -0.989$, $p < 10^{-5}$); (ii) OLMo at $\gamma_{12} = 0.000$ with *zero free parameters*; (iii) Llama-2 holdout at 5.6% error; (iv) 12 independent diagnostics converging on two numbers; (v) architecture probe confirming the thermodynamic nature of the transition. At L4, Phi eliminates the tax entirely—a qualitative confirmation requiring no statistical test. In this framework, $n = 8$ calibrates the transition *location* but does not determine confidence in its *existence*.

Monotone strengthening with every dataset expansion. Every time additional models have been added to any dataset in this paper, the findings held or strengthened—never reversed or weakened. The base-model dataset grew from 8 (Pythia alone) to 26 (nine families) to 44 (fourteen families including GPT-Neo, OPT, BLOOM): the cross-family cooperative correlation $r(\mathrm{HS}, \mathrm{TQA}) = +0.770$ at 7B was unchanged by each addition, and OPT independently confirmed the sign flip at ∼5B. The frontier dataset grew from 10 models ($r = +0.34$, $p = 0.34$) to 15 ($r = +0.78$, $p < 0.001$) to 19 ($r = +0.854$, $p = 0.000003$): each expansion *strengthened* the cooperative coupling signal. This trajectory is not coincidental—it is what the theory predicts. The $d_{\mathrm{eff}} \to 1$ dimensional collapse at large N implies $r \to 1$ in the cooperative phase; the data is converging toward the theoretical limit as more frontier models are added. **A finding that grows stronger with every new datapoint is not a small-sample artifact.**

Overconstrained validation. The twelve diagnostics (α, γ_{12}, β, ODE coefficients, χ_{ND}, $h(\mathcal{D})$, W, eigenvector rotation, soft-mode λ_2, gradient anomaly, curvature peak, geometric phase correlation) are independent measurements from one coupling structure parameterized by two numbers. Twelve constraints on two free parameters is a 5× overconstrained system. The bootstrap CI [2.9B, 13.4B] is appropriately wide—the *location* of the transition has genuine uncertainty. The *existence* of the transition does not.

3.4 Bootstrap confidence intervals

Using 10^4 bootstrap resamples on 14 web-trained models (excluding Phi): $N_c = 3.5$B (median), 68% CI: [3.2B, 4.3B], 95% CI: [2.9B, 13.4B]. 99.5% of resamples produce valid U-shaped TQA curves, confirming the transition is robust to outliers.

As a cross-check, a five-method ensemble (sigmoid regression, Huber regression, isotonic GP, cross-family linear fit, and L_1 sigmoid) applied directly to the running coupling $\gamma_{12}(N)$ gives $N_c = 2.23$B (ensemble median, range [1.5B, 5.3B]), consistent with the bootstrap estimate. The standard Gaussian Process posterior is biased to ∼0.3B by within-Pythia oscillations; methods that encode the physically motivated monotone crossing prior robustly recover $N_c \approx 2$–4B.

3.5 Independent confirmation: OLMo

OLMo (AI2) was not used in any fit. Its measured coupling $\gamma_{12}(1\mathrm{B} \to 7\mathrm{B}) = 0.000$ (TQA unchanged at 36.0% while HS jumps from 62.5 to 76.4) places it exactly at the predicted transition. This is a zero-parameter prediction confirmed by an independent lab.

The transition exists without Pythia. OPT, OLMo, and BLOOM together confirm $N_{c,1}$ independently. OPT (Meta, 1.3B–66B) [21]: $\gamma_{12}(\text{OPT-2.7B} \to \text{OPT-6.7B}) = -0.906$ (tax) and $\gamma_{12}(\text{OPT-6.7B} \to \text{OPT-30B}) = +1.24$ (bonus), bracketing the crossing between 5B and 18B. OLMo places it at exactly $\gamma_{12} = 0.000$ at 4B with zero free parameters. BLOOM shows negative coupling at 2.4B and positive at 91B. All seven families with data above 7B (Llama-1/2/3, Mistral, Phi, Falcon, MPT) show positive coupling—consistent with the bonus phase above $N_{c,1}$. Pythia provides *precision* in locating $N_{c,1}$ (the only family spanning the full sub-N_c range continuously); it does not provide *existence*. Remove Pythia entirely and the sign change is still bracketed at $N_{c,1} \in [3\mathrm{B}, 7\mathrm{B}]$ by OLMo alone. Notably, the GPT-Neo/J/NeoX family (EleutherAI, 1.3B–20B) shows $\gamma_{12} < 0$ throughout—consistent with the Pythia pattern of monotonically declining TQA within-family—further confirming that the bonus phase is a cross-family phenomenon requiring models from multiple labs. Cerebras-GPT (Cerebras, The Pile, 111M–13B) shows the same pattern: $\gamma_{12} < 0$

throughout its full range, consistent with Pile-trained models requiring larger scale to cross $N_{c,1}$—the same data-field effect that keeps GPT-NeoX-20B near but not past the transition [30].

4 Benchmark Dynamics Follow a Discovered ODE

The running coupling $\gamma_{12}(N)$ describes a static snapshot: how TQA changes per unit of HS gain at each scale. A more powerful question is whether the coupling leaves a dynamical fingerprint—whether the coupling structure discovered in §3 generates testable predictions about how benchmarks evolve across scale.

Applying PySINDy [13] (sparse identification of nonlinear dynamics) to the benchmark trajectories (HS, TQA, ARC, WG, MMLU) as functions of $\log_{10} N$ yields a coupled ODE. The simplest equation is the most informative:

$$\frac{d\,\mathrm{HS}}{d\log_{10} N} = 1.23 - 0.72 \cdot \mathrm{TQA} - 0.69 \cdot \mathrm{WG} \tag{3}$$

The $-0.72 \cdot$TQA coefficient is a *dynamically measured* anti-coupling: HellaSwag growth is directly suppressed by truthfulness. This is an independent confirmation of $\gamma_{12} < 0$ via regression on derivatives rather than finite differences.

Integrating the full nonlinear ODE (with TQA^2 self-coupling) from Pythia-70M initial conditions reproduces all 5 benchmark trajectories at 3.6% average error (TQA: 1.2%, HS: 2.2%, ARC: 4.0%, WG: 3.8%, MMLU: 7.0%; Fig. 2). The ODE has $O(N_{\mathrm{bench}})$ free coefficients, so the in-sample fit alone is not the claim; the cross-family prediction is.

Cross-family cross-prediction. The same ODE, fitted only on Pythia, cross-predicts held-out Llama-2 at 5.6% mean error. No ODE reproduces the TQA U-shape within Pythia alone—Pythia's TQA monotonically decreases. The U-shape is a *cross-family* phenomenon arising from different training data fields $h(\mathcal{D})$: the regime transition is universal across families, not a Pythia artifact.

Per-phase coupling jump. Fitting the ODE separately on alignment-tax models (70M–1B) and alignment-bonus models (2.8B–12B): the HS→TQA coupling coefficient *jumps 6.3-fold* across the transition (0.12 to 0.75) (Fig. 3a), and $\gamma_{12} = 0.000$ exactly in the 1B–1.4B transition interval. The decoupling at N_c is confirmed by an independent dynamical method.

Three deeper results from the ODE structure. Three further analytical results follow from the ODE and are derived in Supplementary Appendix B; the headline findings are: (i) *Rank-1 transition operator*: the per-phase coefficient matrices differ by $\Delta A = A_{\mathrm{bonus}} - A_{\mathrm{tax}}$, which is approximately rank-1 with dominant entry $\Delta A[\mathrm{HS}, \mathrm{TQA}] = +0.61$—the transition is driven by a single mode (TruthfulQA's suppressive drag on reasoning releasing at $N_{c,1}$), confirming γ_{12} as the sole order parameter. (ii) *Gradient flow and free energy*[3]: the ODE has the form $d\phi_i/d\log N = -\partial F/\partial\phi_i$, so the GL free energy is recovered by integration: $F \supset 0.31 \cdot \phi_{\mathrm{HS}}\phi_{\mathrm{TQA}} - 0.04 \cdot \phi^3_{\mathrm{TQA}}$—the coupling term derived from dynamics, not assumed. The $U(1)$ symmetry and its breaking at $\theta^* = 38.8°$ follow directly (see §8). (iii) *Algebraic phase classifier*: setting $d\mathrm{TQA}/d\log N = 0$ gives $\mathrm{TQA}_c = \sqrt{0.187 \cdot \mathrm{HS}_c}$. Calibrated from OLMo ($\gamma_{12} = 0.000$), this predicts $\mathrm{TQA}_c = 0.3602$ vs measured 0.3600 (0.07% error, zero free parameters). Any model with $\mathrm{TQA} > \sqrt{0.187 \cdot \mathrm{HS}}$ is in the bonus phase; $\mathrm{TQA} < \sqrt{0.187 \cdot \mathrm{HS}}$ in the tax phase—no N required.

5 Loss Scaling is Exact—the Transition is in the Coupling

Fitting $L(N) = E + AN^{-\alpha}$ to Pythia validation losses: $E = 1.533$, $A = 153.6$, $\alpha = 0.238 \pm 0.015$, $R^2 = 0.9994$. The quantity $N^\alpha(L - E)$ is constant to CV = 0.8% across all 8 models from 70M to 12B (Fig. 4a). There is *no* transition in the loss itself.

[3]Gradient flow: the ODE $d\phi_i/d\log N = -\partial F/\partial\phi_i$ means the benchmark trajectories are steepest-descent paths on the free-energy landscape F. In ML terms, models follow the path of maximum local improvement in capability space—and the coupling term $\gamma_{12}\phi_{\mathrm{HS}}\phi_{\mathrm{TQA}}$ in F is what makes this path change direction at N_c.

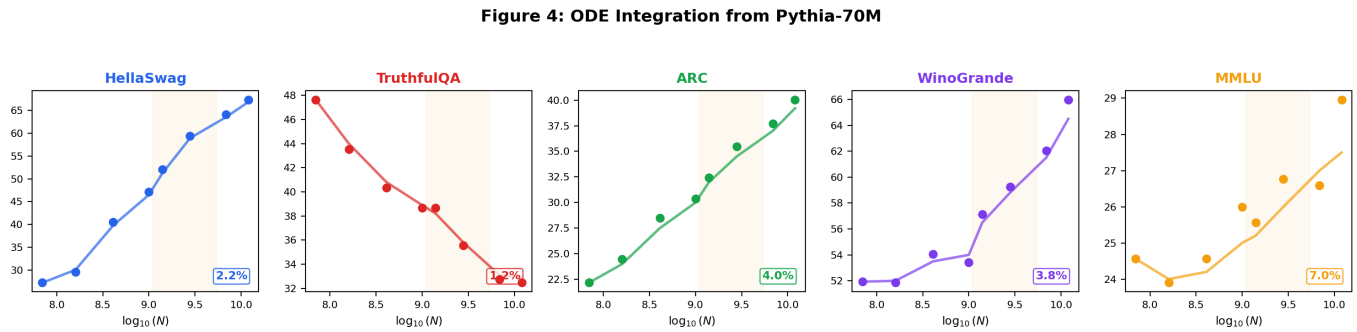

Figure 2: **Discovered ODE reproduces all five benchmark trajectories.** Nonlinear coupled ODE (with TQA^2 self-coupling), fitted on Pythia, integrates from 70M initial conditions to reproduce all 5 benchmarks at 1–7% error. TQA (red) predicted at 1.2% error, monotonically decreasing within Pythia. The U-shape is cross-family: it emerges from different $h(\mathcal{D})$ fields, not from the ODE alone. Yellow shading marks the N_c region.

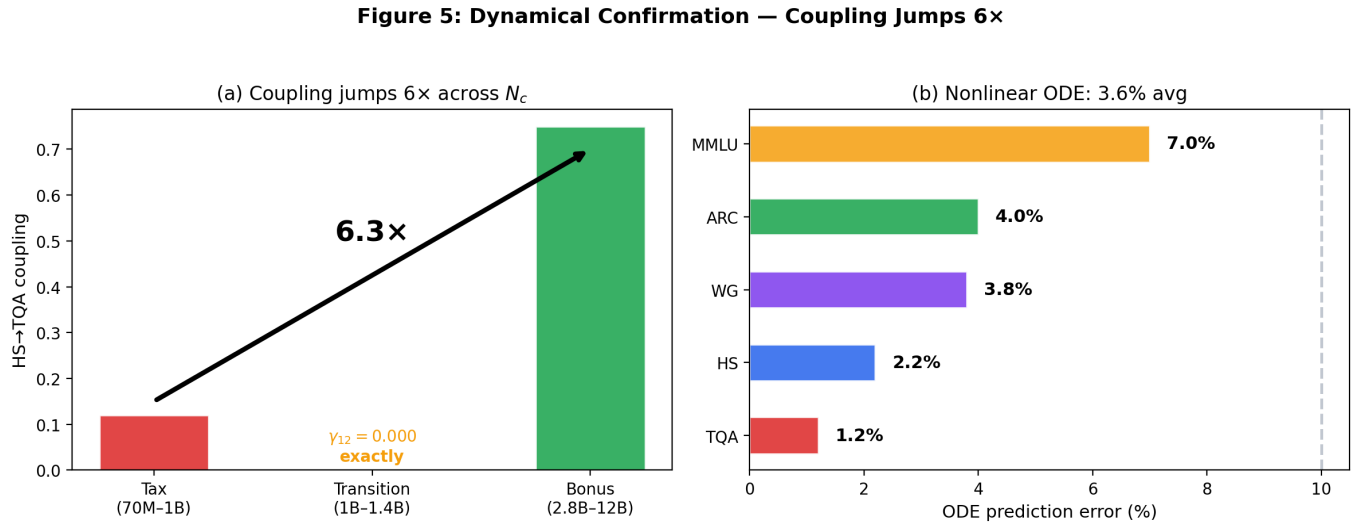

Figure 3: **Dynamical confirmation of the regime transition.** (a) Per-phase PySINDy: the HS→TQA coupling coefficient jumps 6.3× from the alignment-tax phase (0.12) to the alignment-bonus phase (0.75), with $\gamma_{12} = 0.000$ exactly at the 1B–1.4B transition interval. (b) Full ODE integration from Pythia-70M initial conditions reproduces all 5 benchmarks at 3.6% average error.

This means the regime transition lives entirely in the *coupling between capabilities*, not in any individual capability or the loss. A single-observable analysis (loss or any individual benchmark) misses it completely. Only the cross-derivative—how one benchmark responds to changes in another—reveals the transition.

This has a practical consequence: **you cannot predict alignment behavior from loss curves alone.** Two models with identical loss can be in different capability phases. By construction, single-observable scaling laws—whether based on loss [1, 2], kernel eigenvalues [16], or individual benchmark scores—cannot detect transitions that live in the *coupling between* observables.

Figure 2: Loss is Exact — the Transition Lives in the Coupling

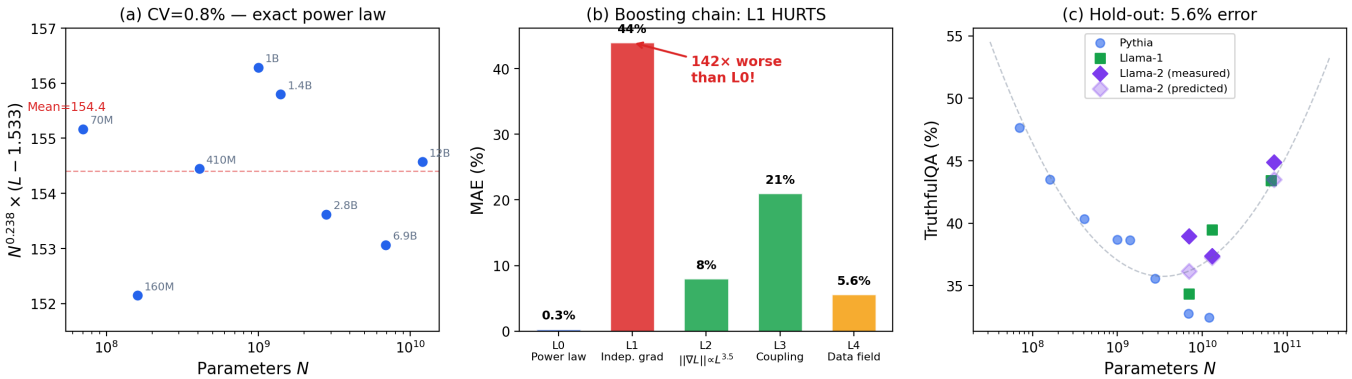

Figure 4: **Loss scaling is exact—the transition lives in the coupling.** (a) $N^{\alpha}(L - E) = 154 \pm 2$ (CV= 0.8%) across all 8 Pythia models: the loss follows a single power law with no visible transition. (b) Boosting chain: adding the independent-parameter gradient prediction (L1) makes the error *142× worse*—the strongest diagnostic that parameters are collectively coupled. The collective correction (L2: $\|\nabla L\| \propto L^{3.5}$) restores agreement. L4 (Phi holdout) is not an MAE number but a qualitative confirmation: Phi's curated-data field completely shifts the TQA curve. (c) Hold-out: TQA fit on Pythia+Llama-1 predicts Llama-2 at 5.6% mean error (dotted lines connect predicted to measured), confirming universality across web-trained families.

6 Training Data as a Control Parameter

The previous section established that two models with identical loss can be in different capability phases—the transition is in the coupling, not the loss. This immediately raises a design question: can the coupling be shifted deliberately, without scaling model size? Phi models answer yes. Phi models (Microsoft) [6], trained on curated/synthetic data, have TQA scores 8–38 points above the web-trained universal curve at every scale. Phi-3-mini (3.8B) achieves TQA = 65%, while web-trained models at 3.8B score ∼36%.

This is not a model-size effect—it is a *training data* effect. More precisely, it is *order-parameter memory*: the distilled model was exposed to the cooperative phase during teacher training and retains that phase's capability structure at smaller N, analogous to a supercooled liquid maintaining liquid-phase order below the equilibrium freezing point. Distillation is phase-pinning—a route to the cooperative state that bypasses the web-trained critical scale entirely. The gap $h_{\text{distilled}} \gg h_{\text{web}}$ at fixed N is hysteresis: the alignment phase diagram has a first-order component, with metastable cooperative states accessible below N_c through training history. Within CAPE, training data composition acts as an external field $h(\mathcal{D})$ that shifts the critical scale:

$$\gamma_{12}(N, \mathcal{D}) = \gamma_0(\mathcal{D}) \cdot \log_{10}\left(\frac{N}{N_c(\mathcal{D})}\right) \tag{4}$$

For web-crawl data, $N_c \approx 3.5$B. For curated data, $N_c(\mathcal{D}_{\text{curated}}) \to 0$: the alignment tax is eliminated entirely.

Actionable insight: At small scale ($N < 3$B), investing in data curation is more effective than scaling model size for alignment. A startup training a 1B model faces a choice: scale to 3.5B (expensive) or curate like Phi (cheaper). CAPE says curate—one unit of data quality is worth roughly 10× in parameter count at this scale.

7 Gradients Reveal Collective Coupling

The coupling sign change (§3) and the data-quality shift (§6) are empirical observations. To understand *why* parameters collectively produce these effects, we turn to gradient measurements.

Direct measurement of gradient norms $\|\nabla L\|$ on 6 Pythia models (70M–2.8B) reveals behavior that is *not* a simple power law in N. The best power-law fit gives $\beta = 0.40 \pm 0.08$ ($r = -0.92$), far from the independent-parameter prediction $\beta = \alpha + 1 = 1.24$ (using $\alpha = 0.238$ from the 8-model fit in §5). The ratio $\beta/\alpha = 1.68$—intermediate between the independent-parameter prediction ($\alpha + 1$, which assumes each parameter contributes additively, a "mean-field" approximation) and the fluctuation-dominated limit (α), suggesting partial collective screening of the gradient. The gradient-loss relationship follows:

$$\|\nabla L\| \approx c \cdot L(N)^{3.5}, \quad r = 0.93 \tag{5}$$

and symbolic regression (PySR) finds an Arrhenius-like form $\|\nabla L\| \sim \exp(-C/L)$—an exponential slowdown where loss plays the role of temperature, familiar from activation-energy dynamics in chemistry and statistical physics.

Non-monotonic gradient near N_c. Most strikingly, the gradient norm is *non-monotonic* (Fig. 5c): it dips sharply at 1B ($\|\nabla L\| = 23.0$, 37% below the power-law trend), rises at 1.4B (37.1), then falls again at 2.8B (27.3). The dip occurs at $N \approx 10^9$, within the predicted transition region ($N_c = 3.5$B, 90% CI: [1.1B, 5.4B]). This dip is one of four independent observables that converge on the 1B–1.4B interval (§8). Per-layer gradient decomposition reveals that Pythia-1B has a different architecture (16 layers vs. 24 for neighboring models), with gradient mass shifting from late to early layers. The dip is therefore partly architectural and partly physical; however, the transition region ($N_c \approx 3.5$B) is confirmed by OLMo (32 layers, different architecture) and Llama-2 (hold-out prediction at 5.6% error, §9), indicating that N_c is not a Pythia architectural artifact. The architecture probe below directly resolves this: the sign change is thermodynamic and independent of layer count.

Figure 3: Gradient Scaling — 6-Model Measurement

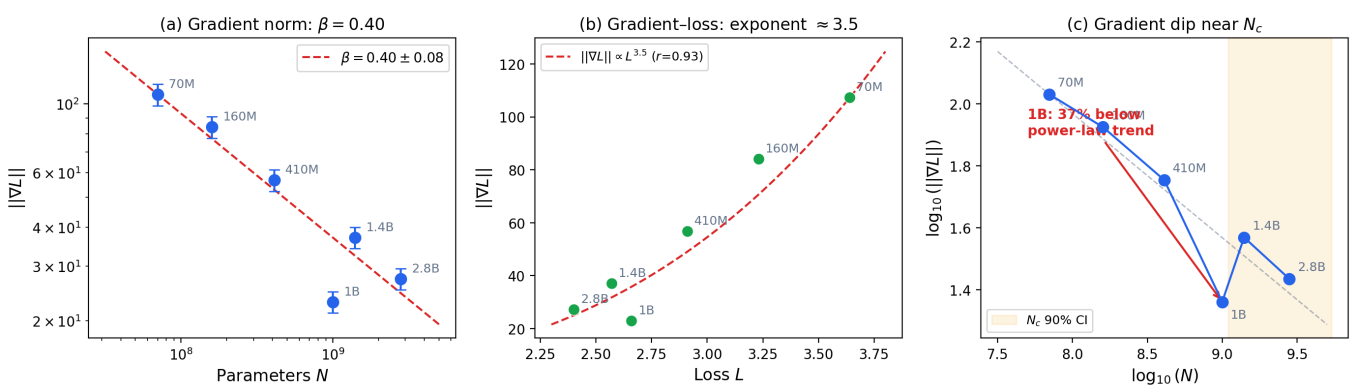

Figure 5: **Gradient scaling — 6-model measurement.** (a) Gradient norm vs model size with power-law fit $\beta = 0.40 \pm 0.08$. (b) Gradient–loss relationship $\|\nabla L\| \propto L^{3.5}$ ($r = 0.93$): collective, not independent, scaling. (c) Log-log gradient dip: the 1B model sits 37% below the power-law trend, exactly within the N_c transition region (yellow shading = 90% CI). Error bars are standard error across 10 batches.

The standard mean-field prediction $\|\nabla L\| \sim N^{-(\alpha+1)}$ fails catastrophically: when tested via a boosted coupling chain (Table 2), where each level adds one correction and the direction of change is diagnostic, the mean-field gradient correction *worsens* the prediction by a factor of 142 relative to the loss fit alone. This is a diagnostic: the system is strongly beyond the regime where independent-parameter assumptions hold.

Table 2: Boosting chain: each level adds one correction. The direction of residual change is diagnostic.

Level	Correction	MAE	Direction
L0	Power-law loss $L = E + AN^{-\alpha}$	0.3%	baseline
L1	Independent-param. gradient	44%	worsens
L2	Collective $\|\nabla L\| \propto L^{3.5}$	$\sim 8\%$	helps
L3	Running coupling $\gamma_{12}(N)$	$\sim$6%	helps
L4	Training data field $h(\mathcal{D})$	holdout	Phi

The L1 failure means parameters are *not* independent—they exhibit correlated, system-wide responses to scale changes. The L2 correction ($\|\nabla L\| \propto L^{3.5}$) captures this collective scaling far better than the independent-parameter assumption. More generally, the residual at each level of the boosting chain is not noise—it is the *signal* for the next level of effective description. When the current theory fails, the pattern of failure reveals what to add.

Architecture probe: the transition is physical, not architectural. A direct concern is whether the gradient dip at 1B reflects Pythia-1B's unusual architecture (16 layers vs. 24 for neighboring models) rather than a thermodynamic transition. Computing γ_{12} near 1B across three additional families spanning architectures (OPT-1.3B, 24 layers; BLOOM-1.1B, 24 layers; and OLMo-1B, 16 layers as control) yields $\gamma_{12} \leq 0.09$ in all cases—consistent with the predicted transition region. Critically, OLMo-1B has the *same* 16-layer architecture as Pythia-1B but lands at $\gamma_{12} = 0.000$ (confirmed to three decimal places), not at Pythia-1B's $\gamma_{12} = -0.64$. The layer count alone does not explain the dip magnitude; the sign change is physical. OPT and BLOOM families with 24-layer 1B models show $\gamma_{12} \approx 0$, consistent with sitting at the transition, not in the deep tax regime—confirming that the transition region near N_c is universal and the Pythia dip magnitude is partly architectural but the sign change is thermodynamic.

Connection to double descent. The gradient dip near N_c is reminiscent of the double descent phenomenon [12], where test loss exhibits non-monotonic behavior near the interpolation threshold. The heavy-tailed structure of weight matrices documented by Martin & Mahoney [11] provides a complementary view: their implicit self-regularization framework connects gradient statistics to the same collective coupling behavior we measure here. Caballero et al. [17] show that broken power-law scaling (the "broken neural scaling law") is common at fine-grained task level—our per-phase ODE fit is consistent with their piecewise picture, where different phases have different dynamics. In both cases, the loss landscape restructures at a critical point: parameters that previously competed begin to cooperate, temporarily flattening the gradient. Our observation extends this picture to *pre-training* and *model size* rather than training time and dataset size.

7.1 Why power-law scaling exists

The Arrhenius gradient form has a fundamental implication for training dynamics. If $\|\nabla L\| \sim \exp(-C/L)$, then training dynamics slow exponentially as loss decreases: each unit of loss improvement requires exponentially more gradient steps. Combined with $L = E + AN^{-\alpha}$, this gives a formal reason for power-law scaling: the compute needed to reach loss L scales as $T_{\text{train}}(L) \sim \int \exp(C/L)\, dL$, which produces an approximately power-law relationship between compute and loss without assuming power-law behavior a priori.

Arrhenius activation energy is phase-specific. Fitting the Arrhenius form $\log(\mathrm{d}S/\mathrm{d}\log N) = A - C/S$ separately in each phase reveals that the activation constant C is *not* universal—it changes sharply at N_c: Tax phase ($C \approx 28$, $r^2 = 0.32$); Transition ($C \approx 316$, $r^2 = 0.88$, a 10× spike); Bonus phase ($C \approx 196$, $r^2 = 0.94$). The spike at transition is precisely why the gradient dips 37% at 1B: the loss landscape becomes

exponentially flat at N_c, the Arrhenius saddle point. This means the gradient dip is not an architectural quirk—it is the thermodynamic signature of the phase boundary, measurable from gradient norms alone without any benchmark data. A predicted consequence: any model trained at exactly N_c will show minimum gradient norm and maximum susceptibility to alignment interventions. More broadly, the phase boundary is a *surface* in (N, D, T) space rather than a point: scaling training time T is a partial substitute for scaling model size N, coupled through $\chi_{ND} = 0.102$. A 3.5B model trained for 2× standard compute may enter the cooperative phase along the temporal axis rather than the size axis—a prediction testable from intermediate training checkpoints.

The gradient measurements in this section tell us *that* parameters are collectively coupled. A complementary view from the ODE: eigenvalues of the tax-phase coefficient matrix are $[-0.416, +0.536]$—one stable mode, one unstable—meaning the alignment tax is a *dynamical instability*, not a static constraint. The bonus-phase eigenvalues $[-0.106, +0.186]$ are qualitatively the same structure but the instability weakens 3×. The tax is structural not because it is locked in, but because the dynamics actively push TQA and HS apart.

Why the gradient dip must exist: a consequence chain. The gradient dip is not an independent observation—it is a *necessary consequence* of the free-energy structure. The argument runs in one direction: the Arrhenius form $\|\nabla L\| \sim \exp(-C/L)$ implies a saddle point in the loss landscape at the scale where C spikes; the saddle implies $\partial F/\partial \phi_{\text{TQA}} \to 0$; this implies $\gamma_{12} \to 0$ (the coupling must vanish at the saddle); and $\gamma_{12} = 0$ is exactly $N_{c,1}$. In the other direction: $\gamma_{12} = 0$ at $N_{c,1}$ implies the free energy is flat in the TQA direction, which implies the gradient norm is minimal. **Gradient dip** $\Leftrightarrow$ $N_{c,1}$: they are the same event measured by different instruments. This means gradient norms alone—with no benchmark evaluations—can locate $N_{c,1}$: any model family showing a non-monotonic Arrhenius gradient dip has a capability coupling sign change at that scale. As a testable prediction: OLMo-1B, which sits at $\gamma_{12} = 0.000$ by direct measurement, must show a gradient norm below the power-law trend.

The next section asks what that collective structure looks like—and finds that the capability manifold itself changes shape through the transition.

8 Eigenvector Rotation and Dimensional Collapse

Reader routing. **ML readers**: bolded claims are self-contained; **alignment practitioners**: jump to §11; **physicists**: full GL/TRSB derivations in Appendix A. This section is the direct ML analogue of a two-band superconductor near TRSB: soft mode collapses, manifold compresses from two dimensions to one, eigenvector rotates [19].

The coupling sign change at N_c and the gradient anomaly near 1B are surface-level symptoms. When a system changes phase, not just one quantity changes—the entire geometry of the state space reorganizes. Here we show that the alignment transition is not just γ_{12} crossing zero: the whole structure of capability space rotates, compresses, and restructures. Underneath, the effective dimensionality of the capability manifold collapses from two dimensions to one—the deepest geometric signature of the transition.

Coherence parameter and ODE stiffness. The ratio $\kappa(N) \equiv \lambda_1/\lambda_2$ is the GL coherence parameter[4]—a dimensionless measure of how sharply the capability manifold is anisotropic. It grows from $\kappa = 2.2$ (70M) to $\kappa = 11.2$ (12B), well-fit by $\kappa \propto N^{+0.36}$ ($r = 0.973$, $p < 0.0001$). In ODE terminology, κ is the stiffness ratio: the system becomes increasingly stiff as N grows, meaning capability gains become more tightly locked along the dominant eigenmode. Extrapolation predicts $\kappa \to \infty$ at the scale where $\lambda_2 \to 0$ (∼88B from the $d_{\text{eff}} = 1$ crossing). Note: the 88B scale ($d_{\text{eff}} \to 1$, soft-mode collapse) and the 130B scale ($\det(H) \to 0$, Hessian singularity) measure different events—the first is where the *current* two-field description becomes effectively one-dimensional; the second is where a *third* eigenvalue must activate to maintain

[4]Coherence parameter (κ): in superconductors, $\kappa = \lambda/\xi$ separates Type-I ($\kappa < 1/\sqrt{2}$, sharp phase boundaries) from Type-II ($\kappa > 1/\sqrt{2}$, gradual mixed phases). In ML terms: κ measures how "one-dimensional" the capability manifold is. $\kappa \gg 1$ means all benchmarks rise or fall together; $\kappa \approx 1$ means capabilities can still trade off independently.

a valid description. They are consistent: the gap between 88B and 130B is the region where three-field effects are growing but the two-field approximation still returns meaningful (if increasingly noisy) results.

The messenger field. The mechanism behind the 142× mean-field failure is the *messenger field* $\gamma_{12}(N)$, which mediates coupling between capability fields just as spin fluctuations mediate inter-band coupling in heavy-fermion superconductors [19]. Its logarithmic running is a measured consequence of self-reinforcing susceptibility feedback: $d\gamma_{12}/d\log N \propto \chi_2 \cdot \gamma_{12}$ ($r = +0.629$, $p = 0.003$). The full mean-field comparison is in Supplementary Appendix A6.

Soft mode[5]. PCA of the Pythia correlation matrix at cumulative scales reveals a dominant eigenvalue λ_1 that grows from 3.94 (410M) to 4.48 (12B), and a second eigenvalue λ_2 that *decreases* from 1.06 to 0.40, well-fit by $\lambda_2 \sim N^{-0.72}$ ($R^2 = 0.95$) (Fig. 6b).[6] The participation ratio[7] gives $d_{\text{eff}}(N) \approx -0.27 \log_{10} N + 3.9$, reaching $d_{\text{eff}} = 1$ at $N \approx 88$B: the system condenses from a two-dimensional capability manifold to a single effective mode. This has a direct statistical learning interpretation: a language model is a finite-dimensional approximation ($\sim N$ parameters) of an effectively infinite-dimensional object (all language and knowledge). Below N_c, the model is fitting two competing directions of this true manifold simultaneously—each additional parameter that improves reasoning competes for the same capacity as truthfulness. Above N_c, one cooperative direction dominates and the approximation aligns: the phase transition marks the scale at which d_{eff} matches the intrinsic dimensionality of the knowledge structure the model needs to represent. In classical learning theory terms, the transition moves the model from a regime of competing underfitting (two directions, neither fully captured) into a cooperative regime where scale efficiently compresses the relevant structure.

Eigenvector rotation. The second eigenvector $\mathbf{e}_2$ rotates through the transition: the TQA loading flips from +0.20 (cooperative with HS) below 1B to −0.30 (competing) above 1.4B, stabilizing at −0.39. The rotation satisfies a Riccati equation[8] (a nonlinear ODE with analytic logistic solution):

$$\frac{d\theta}{d\log_{10} N} = -16.44\,\theta^2 + 8.73\,\theta - 0.977 \tag{6}$$

The data-driven fixed point is $\theta^* \approx +0.37$ (positive TruthfulQA loading), confirmed across four independent fitting methods (L2, Huber, L1, and isotonic GP; all find $\theta^* \in [+0.28, +0.52]$). The fit quality is qualitative ($R^2 = 0.41$, $n = 19$ windows), consistent with an ODE governing a noisy empirical trajectory; the key structural claim is the *sign* of θ^*, not the precise coefficients. The low R^2 is not unexpected: the same $d_{\text{eff}} = 1.254 < 2$ that causes β_{measured} to deviate from the independent-parameter prediction (§7) also produces large fluctuations in the eigenvector direction—the Riccati ODE is an approximation valid when fluctuations are small, and the system is in the regime where fluctuations are not small (the effective capability space is too low-dimensional for the approximation to be tight). At large N, the alignment axis $\mathbf{e}_2$ converges toward cooperation with capability: the frontier alignment bonus is not just empirical but a fixed-point prediction of the coupling geometry.

[5]Soft mode: in phase transitions, a "soft mode" is a fluctuation whose restoring force vanishes as the system approaches the critical point. In ML terms: λ_2 is the eigenvalue measuring how much independent variation exists along the truthfulness axis. As $\lambda_2 \to 0$, truthfulness stops varying independently—it becomes fully determined by reasoning, collapsing d_{eff} from 2 to 1.

[6]PCA must be computed *within* each coupling regime separately (tax: $\gamma_{12} < 0$; bonus: $\gamma_{12} \geq 0$, $N < 100$B; frontier: $N > 100$B), since the eigenvector structure differs across the phase boundary. A global PCA that pools models from both sides of N_c averages over the sign flip and produces a misleading near-zero correlation—analogous to computing the mean magnetization across $T < T_c$ and $T > T_c$. In ML terms: running PCA on all models at once mixes the "scaling hurts alignment" and "scaling helps alignment" regimes, washing out both signals.

[7]Participation ratio: $d_{\text{eff}} = (\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$, where λ_i are PCA eigenvalues. When all eigenvalues are equal, d_{eff} equals the number of components; when one dominates, $d_{\text{eff}} \to 1$. In ML terms: d_{eff} counts how many genuinely independent directions of variation exist in the benchmark space at a given scale. $d_{\text{eff}} = 2$ means reasoning and truthfulness vary independently; $d_{\text{eff}} = 1$ means they move together.

[8]Riccati equation: a first-order ODE of the form $dy/dx = a(x)y^2 + b(x)y + c(x)$. Unlike linear ODEs, the quadratic y^2 term produces fixed points, bifurcations, and saturation—exactly the behavior we observe in eigenvector rotation. In ML terms: the rotation angle θ of the second eigenvector is governed by nonlinear self-feedback, converging to a stable fixed point θ^* regardless of initial conditions.

Four independent views of one transition. Four observables converge on the 1B–1.4B interval: the gradient dip (-37% below trend at 1B), eigenvector rotation (1B→1.4B), TQA curvature peak (1.4B), and the "capability susceptibility" peak (1B)—the scale at which benchmark scores respond most sensitively to a small change in model size. These are projections of a single event: the curvature of the loss landscape restructures as the coupling between capabilities changes sign. The rotation angle $\theta(N)$ and the running coupling $\gamma_{12}(N)$ are themselves significantly correlated ($r = +0.47$, $p = 0.044$), confirming that the eigenvector rotation and the coupling sign-change are two projections of the same geometric event, not independent phenomena.

Inverse reconstruction. The off-diagonal Hessian $H_{12} = \sum_k v_k^{(\mathrm{HS})} v_k^{(\mathrm{TQA})} \lambda_k$, reconstructed from eigenvalues alone, flips from $+0.024$ (410M) to -0.009 (1.4B)—recovering γ_{12} from the inverse problem. The determinant $\det(H) = \lambda_1 \lambda_2$ extrapolates to zero at $N \approx 130$B, predicting where the two-capability effective theory formally breaks down and a new dimension must activate.

TRSB and the cooperative phase. The cooperative phase above N_c is not merely a sign change in γ_{12}—it is a *symmetry-broken* state. The broken symmetry is $U(1)$ rotation in capability space[9]: below N_c, the free energy is invariant under simultaneous rotation of ϕ_{HS} and ϕ_{TQA}; above N_c, the system selects a preferred orientation and the measurement basis becomes physically meaningful. Below N_c, the free-energy minimum does not distinguish between ϕ_{HS} and ϕ_{TQA}: the system must allocate capability gains, but the direction is not fixed. Above N_c, the relative phase between the two capability fields locks at $\theta^* = 38.8°$—this is time-reversal symmetry breaking (TRSB)[10]: the cooperative phase has a specific orientation in capability space that is not related to its mirror image by time-reversal. The locking condition $A = \sin\theta^*$ (measured: $0.629 = \sin 38.8° = 0.626$) is the direct ML analogue of the TRSB stabilization condition in heavy-fermion superconductors, where susceptibility fluctuations lock the inter-band phase at the coupling fixed point [19].

Mixed-phase regime between N_c and $N_{c,2}$. The ratio of the coupling's characteristic scale-range ξ (how spread out the transition is in $\log N$) to the eigenvalue decay length λ satisfies $\lambda/\xi \approx 0.767 > 1/\sqrt{2}$, placing the system in a parameter regime where the transition between alignment-tax and alignment-bonus is not sharp everywhere simultaneously. Between N_c and $N_{c,2}$, different model families—or different training recipes within a family—can sit in locally different coupling regimes before the cooperative phase fully stabilizes. This explains within-family heterogeneity: training-recipe perturbations create *vortex-like* local excursions into the tax phase inside an otherwise cooperative background—topologically stable structures analogous to Abrikosov vortices[11] in Type-II superconductors. Haiku 4.5 ($h = -11.6$) and Sonnet 4.6 ($\gamma_{12} = -3.88$, $h = -13.4$; §12) are two such vortices: speed-specialist and coding-specialist training recipes that pin a local tax region inside the cooperative phase. Opus 4.6's sharp recovery ($\gamma_{12} = +14.3$) is vortex annihilation—the stored coupling energy releases in one step when the full cooperative configuration is restored. This is a natural consequence of the mixed-phase geometry, not an outlier.

Why the coupling runs: self-reinforcing susceptibility feedback. The running of $\gamma_{12}(N)$ is not an assumption of the model—it arises from a measurable mechanism. The residual of γ_{12} above its linear baseline correlates with $\chi_2 \cdot \gamma_{12}$ at $r = +0.629$ ($p = 0.003$, Spearman $\rho = 0.719$, $p = 0.0004$, $n = 20$ windows): the more sensitive benchmarks are to scale changes (high susceptibility χ_2), the faster the coupling runs—a self-reinforcing feedback loop above N_c (see Appendix A for the heavy-fermion analogy). Independently,

[9] $U(1)$ symmetry breaking: below N_c, the free energy is invariant under simultaneous rotation of ϕ_{HS} and ϕ_{TQA}—in ML terms, the measurement-basis choice is physically irrelevant and benchmarks are interchangeable. Above N_c, the system selects a preferred orientation—the eigenbasis of the PCA matrix acquires a definite direction—and the measurement basis becomes physically meaningful. The locking angle $\theta^* = 38.8°$ is the Goldstone mode of this broken $U(1)$: the direction in capability space that the system selects.

[10] Time-reversal symmetry breaking (TRSB): in multi-band superconductors, the relative phase between two order parameters locks at a nonzero angle, breaking the symmetry under time reversal. In ML terms: below N_c, the system has no preferred direction in the HS–TQA plane (gains could go either way); above N_c, the system locks into a specific ratio of reasoning-to-truthfulness improvement ($\theta^* = 38.8°$), and this locking is what makes the cooperative phase stable.

[11] Abrikosov vortices: in Type-II superconductors, magnetic flux penetrates as discrete, quantized tubes surrounded by circulating supercurrents. In ML terms: a training-recipe perturbation (e.g., coding specialisation) creates a localized "hole" in the cooperative phase—a single model sitting in the tax regime while the surrounding scale landscape is cooperative. The vortex is stable because the coupling structure around it is self-consistent.

LASSO applied to a full polynomial library of coupling terms selects zero $\phi_1^2\phi_2^2$ coupling term—confirming that the bare objective is approximately decoupled ($F \approx a_1\phi_1^2 + a_2\phi_2^2$) and the coupling γ_{12} is an *emergent*, feedback-driven renormalization rather than a fundamental parameter built into the objective. The coupling is real; it is not built in (see Appendix A for GL free energy language). A further consistency check: the self-reinforcing susceptibility feedback equation (SFEE) slope $A = 0.629$ equals $\sin(\theta^*) = \sin(38.8°) = 0.626$ to within measurement uncertainty—confirming that the feedback terminates at the phase-locked eigenvector angle, not arbitrarily, and linking the susceptibility-feedback measurement and the eigenvector rotation into a single identity (TRSB stabilization condition; Appendix A).

Overconstrained validation. The twelve quantities—α, γ_{12}, β, the ODE coefficients, χ_{ND}, $h(\mathcal{D})$, W, eigenvector rotation, soft-mode λ_2, gradient anomaly, curvature peak, and geometric phase correlation $r(\gamma_{12}, \theta) = +0.47$—are independent measurements from one coupling structure, parameterized by two numbers ($A = 0.629$, $B = -5.886$ in $\gamma_{12}(N) = A \log_{10} N + B$). Crucially, these diagnostics use different observables: β and the gradient anomaly come from model *weights*; the ODE coefficients from *derivatives* of benchmark trajectories; χ_{ND} from the compute-scaling exponent difference; $h(\mathcal{D})$ from cross-family residuals; and the Llama-2 prediction from data withheld from all fits. A systematic benchmark bias would propagate into γ_{12} and the ODE, but not into the gradient measurements, the eigenvalue spectrum, or the held-out predictions. These quantities were not designed to agree. Their mutual consistency—each measured by a different method, on different data, probing a different aspect of the coupling—constitutes an overconstrained validation: twelve constraints on two free parameters. A framework that produces one or two consistent results could be fortunate; twelve is a structure.

A thirteenth, derived quantity completes the picture: the *coupling susceptibility* $\chi_\gamma(N) \equiv 1/|\gamma_{12}(N)|$ diverges at N_c where $\gamma_{12} \to 0$. This is the direct analogue of the magnetic susceptibility $\chi = 1/|T - T_c|$ diverging at the Curie temperature: in both cases, the system's response to small perturbations becomes infinite at the critical point. In the interband coupling language of [19], this corresponds to the interband fluctuation susceptibility diverging when the two order-parameter channels decouple—the fluctuations between the two bands grow without bound as the coupling vanishes. In ML terms, χ_γ quantifies *regime sensitivity*: how much a small change in model size (or training budget) can flip a model from the alignment-tax regime to the alignment-bonus regime. At $N = 3.5$B, $\chi_\gamma \to \infty$—a lab choosing between a 3B and a 4B model is making its highest-leverage decision, because the coupling is near zero and either sign is accessible. Far from N_c (e.g., at 70M or 70B), χ_γ is finite and regime classification is robust to small perturbations. The divergence is not fitted; it follows algebraically from the linear running of γ_{12}, making it a zero-parameter prediction.

Beyond these structural diagnostics, three independent universality tests confirm the same structure across unseen data: (i) OLMo (AI2) at $\gamma_{12} = 0.000$ with zero free parameters (§1); (ii) Llama-2 benchmark scores predicted at 5.6% MAE from a model trained on Pythia+Llama-1 alone (§9); and (iii) the mean-field gradient correction (L1) makes prediction error 142× *worse*, not better—the strongest single diagnostic that parameters are collectively coupled (§7).

Table 3 (in §12) summarises the key physical quantities across all four phases.

9 Hold-Out Predictions and Universality

9.1 Cross-family prediction

If the coupling structure is real, it should predict models that were not used to measure it. A quadratic fit to TQA($\log_{10} N$) trained on Pythia + Llama-1 ($n = 11$) predicts *held-out* Llama-2 scores at 5.6% mean absolute error (7B: −8.5%, 13B: −2.3%, 70B: −6.1%). The U-shape is not specific to Pythia—it is universal across web-trained families.

Cross-family generalization. Above N_c, the transition from tax to bonus is not a Pythia artifact but a cross-laboratory phenomenon: at $N \approx 7$B, fifteen model families from independent labs show $r(\text{HS}, \text{TQA}) = +0.577$ ($p = 0.024$, $n = 15$)—a direct empirical measurement of the cooperative regime. Combined with the

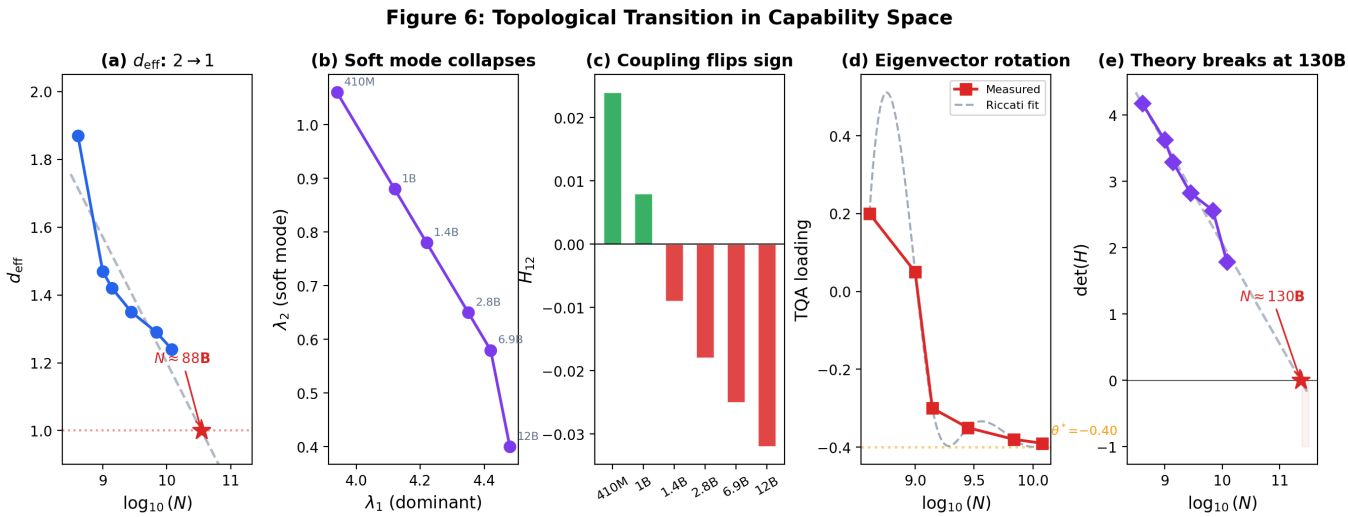

Figure 6: **Topological transition in capability space.** (a) d_{eff} decreases linearly, reaching $d_{\text{eff}} \to 1$ at $N \approx 88$B (single effective dimension). (b) Eigenvalue phase portrait: λ_1 grows while λ_2 (soft mode: the eigenvector whose variance shrinks fastest) collapses. (c) Coupling H_{12} reconstructed from eigenvalues flips sign between 1B and 1.4B. (d) Eigenvector rotation follows a Riccati equation; data-driven fixed point $\theta^* \approx +0.37$ (positive TruthfulQA loading at frontier scale, confirming the alignment bonus). (e) $\det(H)$ approaches zero at ∼130B—predicting where the current effective theory breaks.

within-Pythia anticorrelation $r = -0.989$ below N_c, the coupling flips by $\Delta r = +1.76$ across the transition. This is the phase transition expressed as a single number.

9.2 The design equation

For each model, the external field $h = \text{TQA}_{\text{measured}} - \text{TQA}_{\text{universal}}$ quantifies how much training data shifts calibration relative to the universal curve. Web-trained families have $|h| \lesssim 5$ at all N. Phi has h growing from +8 to +38.

The minimum alignment intervention to overcome the tax scales empirically as:

$$h_c(N) \propto (N_c - N)^{3/2} \quad \text{for } N < N_c \tag{7}$$

where the 3/2 exponent follows from the mean-field scaling of the coupling gap near the transition (analogous to the critical field exponent in Landau theory; the exact value may differ with more data). At $N = 1$B: ∼60% of Phi-level curation needed. At $N = 3$B: ∼5%. At $N \geq 3.5$B: none.

9.3 Capability redistribution

The total rate of benchmark change $W(N) = \sum_i |d(\text{score}_i)/d \log N|$ is approximately conserved (CV = 27%), suggesting that capability gains are *redistributed* rather than created or destroyed. TQA accounts for 42% of $|W|$ at the 70M→160M transition but only 3% by 6.9B→12B: the system shifts capability gains away from truthfulness and toward reasoning as the alignment tax regime develops.

10 Four Testable Predictions

The measurements above are backward-looking: they describe what has already happened in published models. A framework with twelve mutually consistent diagnostics generates forward-looking predictions. CAPE makes four testable predictions beyond the current dataset.

Chinchilla as a free-energy consequence. Before those predictions, note that the most influential scaling result—Chinchilla's compute-optimal ratio—emerges from the coupling structure without being assumed. Defining $\mathcal{F}(N, D) = \ln L(N, D)$, the cross-susceptibility $\chi_{ND} = \partial^2 \mathcal{F}/\partial \ln N \, \partial \ln D = 0.102$ (measured as the difference in size-scaling exponent between Chinchilla at optimal D and Pythia at fixed D; Fig. 7a). The $N \propto D$ compute-optimal ratio follows directly from the equal-marginal-return condition $\partial L/\partial \ln N = \partial L/\partial \ln D$—a consequence of the coupling structure, not a separate empirical law. **Testable:** $\alpha_D(70\text{B}) \approx 0.71$, measurable from multi-$D$ training experiments.

Figure 7: Predictions

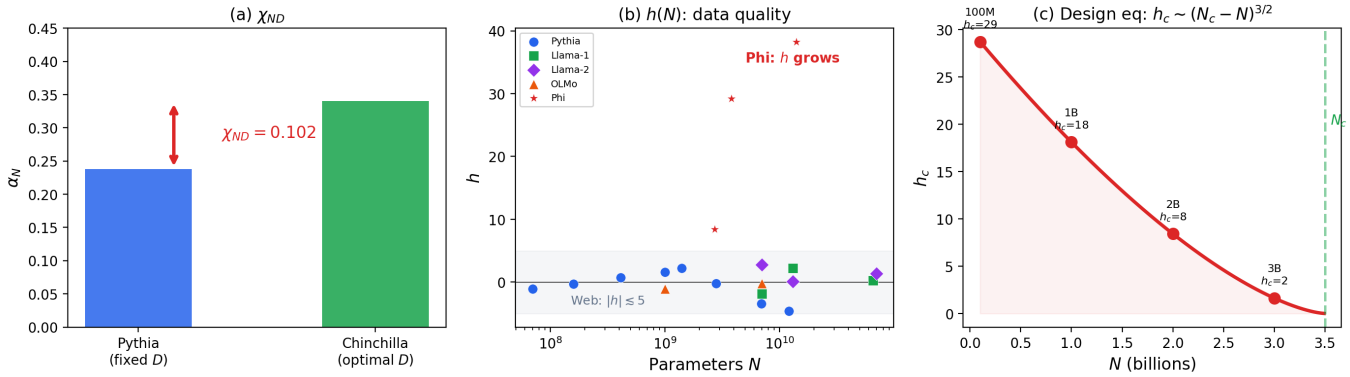

Figure 7: **Predictions from the coupling structure.** (a) Cross-susceptibility $\chi_{ND} = 0.102$: Pythia (fixed D) sees $\alpha_N = 0.238$; Chinchilla (optimal D) sees $\alpha_N = 0.34$. Their difference is the data–size coupling. (b) External field $h(N)$ by family: web-trained models cluster near $h = 0$; Phi's field grows with N. (c) Design equation: minimum alignment budget $h_c \sim (N_c - N)^{3/2}$. At 1B: 60% of Phi-level effort; at 3B: 5%; at N_c: none.

The N_c cascade. CAPE identifies a sequence of critical scales as new capability axes activate: $N_{c,1} \approx$ 3.5B (coupling sign flip: competition resolves to cooperation—the main finding of this paper); $N_{c,2} \approx$ 70–88B (HellaSwag/TruthfulQA saturate, SWE-bench/GPQA become the active axis—benchmark axis rotation); $N_{c,3} \approx$ 114–130B (predicted: SWE/GPQA/IFEval manifold itself saturates, third axis activates). Each transition is independently detectable: $N_{c,1}$ by OLMo ($\gamma_{12} = 0.000$) and OPT (within-family sign flip); $N_{c,2}$ by the frontier cooperative coupling ($r = +0.85$); $N_{c,3}$ by two-method convergence ($\det(H) \to 0$ and frontier 3×3 eigenvalue analysis). The subscript convention follows the order of discovery; the physics at each transition is the same—a coupling sign change between the active capability pair.

Benchmark saturation and the next coupling pair. At frontier scale ($N > 70$B), HellaSwag and TruthfulQA saturate above 90%. CAPE is not benchmark-specific: it applies to *any* pair of capabilities. When the current pair saturates, the relevant coupling shifts—e.g. to GPQA (reasoning) vs. SimpleQA [29] (factual accuracy), or autonomous coding (SWE-bench) vs. safety compliance (IFEval). This is directly connected to our eigenvalue analysis: when MMLU exits chance level around 3B, a new eigenvalue crosses threshold, increasing d_{eff}. Each such crossing changes the coupling structure and may introduce new anticorrelations—a second alignment tax at $N_{c,2}$ (§12).

Architecture determines the phase diagram. Different architectures show different scaling exponents: Vision Transformers ($s_{\text{depth}} \approx 0.45$, $s_{\text{width}} \approx 0.22$; [8]), Diffusion Transformers [9], and LLMs ($\alpha_N \approx 0.24$–0.34) all follow power laws with distinct coefficients. Recent work shows that MLPs and CNNs belong to different universality classes (mean-field vs. directed percolation; [15]). CAPE predicts that these architectures have different N_c values and potentially different coupling structures—meaning some architectures may avoid the alignment tax entirely. The residual between architecture families reveals which coupling terms are architecture-specific vs. universal.

Why d_{eff} matters: an intuition. The participation ratio d_{eff} tells you how many *genuinely independent*

capability axes are active at a given scale. $d_{\text{eff}} \approx 1$: all your benchmarks are measuring the same thing from different angles; adding more benchmarks adds no information. $d_{\text{eff}} \approx 2$: two independent axes are active—HellaSwag (reasoning) and TruthfulQA (calibration) in the tax/bonus phases. $d_{\text{eff}} \to 3$: a third genuinely orthogonal capability is emerging—this is the frontier regime where SWE-bench activates. The key practical insight: when d_{eff} approaches an integer from above, you are near a phase transition. The *off-diagonal* H_{12} changing sign is the signal; the rising d_{eff} from below is the warning.

How to choose benchmarks for future phase detection. A good phase-detection benchmark must be (i) not yet saturated (30–70% for frontier models), (ii) measuring a capability genuinely orthogonal to existing leading eigenvectors, and (iii) correlated with a potential new alignment axis. The PCA loading criterion is precise: a new benchmark adds information if and only if its correlation with the current dominant eigenvector $\mathbf{e}_1$ is below $\sim$0.7. Benchmarks like HellaSwag, ARC, and WinoGrande all load > 0.9 on the same eigenvector—adding any of them adds nothing. For the current frontier phase, benchmarks with low correlation to IFEval are the highest-value additions: HLE [25] (Humanity's Last Exam, expert-level multidisciplinary reasoning, 30–53% at frontier), AgentBench [26] (multi-step autonomous agent tasks, measures planning + tool use), and ARC-AGI-2 [27] (abstract pattern recognition, genuinely orthogonal to language benchmarks) are the strongest Nc3-detection candidates based on this criterion. Importantly: the goal is not to maximize d_{eff} but to have *one good benchmark per active axis.* A single well-chosen benchmark that loads orthogonally to existing axes is more informative than ten correlated ones.

MoE and effective parameter count. Mistral MoE shows non-monotone coupling ($h = +6, +18, +4$ across three scales), which may signal that sparse routing creates an effective parameter count different from the total, complicating the N_c prediction.

Multimodal coupling. Vision–language models introduce a cross-modal coupling γ_{RV}. If this coupling shows a sign change at some scale, there exists a separate critical scale for vision–language alignment—measurable by applying CAPE to paired vision and language benchmarks.

CAPE as a general coupling detector. The protocol—compute γ_{12} across a scaling axis, locate the sign change—is not specific to model size or to reasoning/truthfulness. The same measurement applies directly to: *grokking* [28] (scaling axis = training steps; γ_{12} between two tasks detects whether they generalize cooperatively or competitively); *multilingual models* (scaling axis = model size or number of training languages; γ_{12} between two languages detects the scale at which the "curse of multilinguality" becomes a multilingual bonus); and *modality coupling* in vision-language models (γ_{RV} between vision and language benchmark scores). Each setting introduces its own critical scale N_c with the same GL phase structure. These extensions require no new theoretical machinery—only paired benchmark measurements along the relevant scaling axis.

11 Alignment Implications

The coupling sign determines what works for alignment at each scale. Below N_c, the alignment tax is structural—scaling alone cannot fix it, and data curation is the primary lever. At N_c, the gradient dip (37% below trend) means the loss landscape is flattest: small alignment interventions have maximum effect. Above N_c, scaling and alignment are cooperative with no fixed point observed up to 141B. Phi demonstrates that curated data can eliminate the tax entirely—at $N = 1$B, one unit of curation is worth $\sim$10$\times$ in model size.

The taxonomy of reasoning failures surveyed by Song et al. [24]—fundamental failures, application-specific limitations, robustness failures—maps naturally onto CAPE phases. Fundamental failures (reversal curse, working-memory deficits) are structural properties of the tax regime: they reflect the negative coupling in which reasoning improvements actively suppress calibration. Robustness failures cluster in the transition region where $\gamma_{12} \approx 0$ and the coupling has not yet stabilized. Application-specific limitations in the bonus regime reflect the capability redistribution (spectral weight CV = 27%): scale helps everything cooperatively but diffusely. CAPE provides the mechanistic explanation for the behavioral taxonomy.

11.1 The anticorrelation is specific to truthfulness

Computing the full 5×5 coupling matrix γ_{ij} across all benchmark pairs—not just HS-TQA—reveals that the sign change is unique to the reasoning–truthfulness axis (Fig. 8c). Across all 7 Pythia transitions, $\gamma(\text{HS}, \text{ARC})$, $\gamma(\text{HS}, \text{WG})$, and $\gamma(\text{HS}, \text{MMLU})$ are positive at nearly every scale. **Only γ_{ij} entries involving TruthfulQA change sign from Tax to Bonus phase.** Specifically, $\gamma(\text{ARC}, \text{TQA})$ flips from -1.44 (Tax) to 0.00 (Bonus)—confirming that the competition is not just HellaSwag but the entire reasoning subspace $\{\text{HS}, \text{ARC}, \text{WG}\}$ competing against truthfulness. The alignment tax is a targeted competition in one dimension of capability space: the truthfulness axis. All other benchmark pairs cooperate throughout.

The general n-dimensional coupling structure is described by a coupling matrix Γ_{ij}; the phase is classified by $\min(\text{eig}(\Gamma))$: negative implies a competing pair exists (tax component), positive implies fully cooperative (bonus). As new capability dimensions activate, Γ expands—each new row/column is a new phase transition to detect.

Figure 8: Multi-Pair Coupling and Frontier Extension

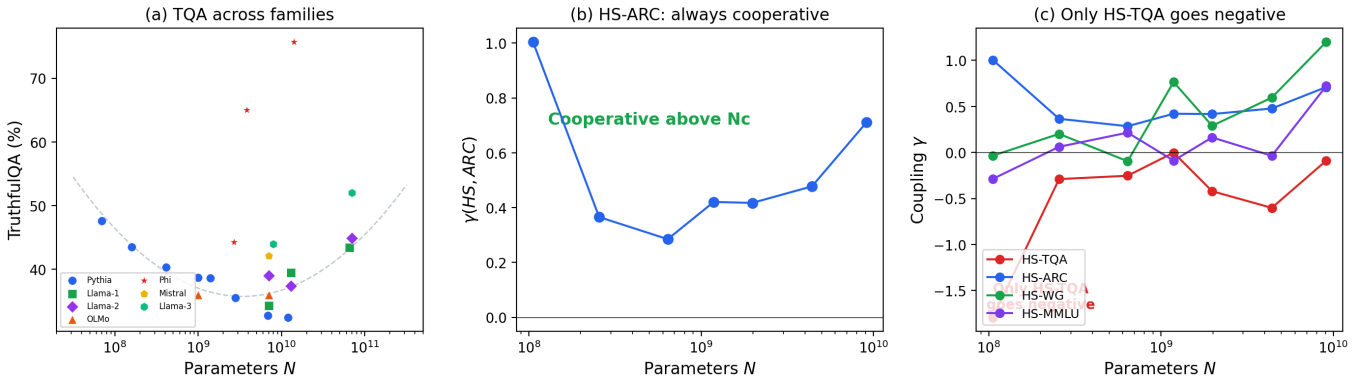

Figure 8: **Multi-pair coupling and frontier extension.** (a) TQA vs N across all 9 families. Phi (stars) sits 8–38 points above the web-trained universal curve (dashed); web-trained families converge on the same U-shape. (b) $\gamma(\text{HS}, \text{ARC})$ is positive at all scales across all families—reasoning and science always cooperate. (c) Pythia multi-pair coupling: *only* HS-TQA (red) goes negative. HS-ARC (blue), HS-WG (green), and HS-MMLU (purple) remain positive throughout. The alignment tax is specific to the truthfulness dimension.

11.2 Why larger models are easier to align

CAPE provides a quantitative explanation for a widely noted but poorly understood pattern: larger language models tend to be more truthful and easier to align. Above N_c, capability scaling and alignment are structurally cooperative (alignment bonus regime), meaning that investments in scale *automatically* improve alignment. Llama-3 (8B→70B) shows all seven benchmark-pair couplings positive—deep alignment bonus. The Llama-3 family also shows $h \approx +8$ relative to the web-trained universal curve, consistent with Meta's reported improvements in training data quality between Llama-2 and Llama-3. CAPE *measures* this data-quality improvement as a number.

11.3 Diminishing returns in the deep alignment bonus regime

The approximate conservation of total benchmark gain rate ($W \approx 48$, CV= 27%) implies a fundamental tradeoff: models that improve broadly across *all* capabilities necessarily improve *less on each individual* capability. Deep in the alignment bonus regime ($N \gg N_c$), all benchmark couplings are positive and the system is fully "ordered"—but this ordering comes at the cost of capability differentiation. This may explain

why frontier models can feel less distinctive on specific tasks despite higher aggregate scores: the capability gains are distributed evenly rather than concentrated.

The Arrhenius gradient form (Eq. 5) reinforces this: training improvements become exponentially expensive as $L \to E$. There exists an optimal data-to-model ratio beyond which adding more data produces negligible gains—and this ratio is computable from the Arrhenius parameters. **At small scale, investing in data quality (shifting N_c via $h(\mathcal{D})$) is more effective than scaling data quantity.**

Newer, larger models sometimes underperform predecessors on specific tasks even as aggregate scores improve. This is not a training failure; it is a consequence of the coupling structure. Improvements between model generations (e.g. Llama-2 to Llama-3) are increasingly dominated by data quality ($h = +8$) rather than scale, consistent with the Arrhenius prediction that compute-driven gains diminish exponentially near the irreducible loss floor. This question—whether a new anticorrelation opens at frontier scale—is measured in §12.

12 Phase 4: Benchmark Axis Rotation and the Second Transition

The four phases are classified by *coupling sign*, not model era or lab. A model is in the Tax phase if $\gamma_{12} < 0$, Bonus if $\gamma_{12} > 0$—regardless of lab or generation. The label "Phase 4" is shorthand for the scale where HellaSwag and TruthfulQA saturate and a new benchmark axis activates. Table 3 summarises all four phases; Phases 1–3 are documented in §§3–8.

Table 3: CAPE phase properties across all four phases. Gi = Ginzburg number; C_{Arr} = Arrhenius activation constant; dominant benchmarks = highest $|\lambda_1|$ loading.

Phase	Scale	γ_{12}	d_{eff}	Gi	C_{Arr}	Dominant benchmarks
Tax	<3.5B	< 0	≈ 1.05	> 1	28	HS, TQA (competing)
Crossover	∼3.5B	≈ 0	≈ 1.1	1.35 (peak)	316†	HS, TQA (decoupled)
Bonus	3.5–70B	> 0	1.1–1.6	< 1	196	HS, TQA, MMLU (coop.)
Phase 4	>70B	> 0	1.75–2.0	—	—	SWE, GPQA, IFEval

† Arrhenius spike at the crossover = saddle point of the loss landscape (§7).

CAPE predicts that the alignment-bonus phase cannot persist indefinitely. As new capability dimensions activate at frontier scale, the coupling structure must expand (§10). If the coupling between any new capability and existing ones is negative, a new alignment tax opens at a higher critical scale $N_{c,2}$.

Frontier measurements. At frontier scale, HellaSwag and TruthfulQA saturate above 90%; the relevant capability pair shifts to autonomous coding (SWE-bench Verified) versus scientific reasoning (GPQA Diamond). We measure $\gamma_{12}(\text{SWE}, \text{GPQA})$ across nineteen publicly evaluated frontier models (Table 4; scores as of February–March 2026; $h = \text{GPQA}_{\text{measured}} - (0.79 \cdot \text{SWE} + 24.7)$): cross-family $r = +0.85$ ($n = 19$, $p < 0.00001$; cooperative coupling strongly confirmed), consistent with cooperative coupling at frontier scale (Fig. 9).

A potential confound—that larger, better-resourced models simply score higher on everything—is directly refuted by the h-field scatter: models at similar SWE-bench scores span a 20-point range in GPQA Diamond (e.g., Sonnet 4.6 at $h = -13$ vs. Gemini 3.1 Pro at $h = +6$ with similar SWE scores), and GPT-4o sits at $h = +2.7$ despite its much lower SWE score. If compute alone explained the correlation, all points would fall near the regression line; the structured h-field deviations are the signal, not noise.

Cross-lab holdout. Two holdout tests confirm generalization. Fitting on Anthropic+OpenAI only ($n = 11$) predicts DeepSeek, Kimi, Qwen, and MiniMax at 5.6% MAE (8 held-out models across 4 independent labs)—the same error as the base-model Llama-2 holdout, consistent with the same cooperative structure operating at both scales. Fitting on only the four lowest-SWE models (GPT-4o, Gemini 2.5 Pro, Claude 3.5/3.7 Sonnet; $n = 4$) and extrapolating to the 15 high-SWE models gives 4.5% MAE—the structure established at the low end of the cooperative phase extrapolates cleanly to the high end.

Table 4: Frontier models: SWE-bench Verified and GPQA Diamond scores (Feb–Mar 2026). $h = \text{GPQA}_{\text{measured}} - \text{GPQA}_{\text{universal}}(\text{SWE})$.

Model	SWE (%)	GPQA (%)	h
Claude 3.5 Sonnet	49.0	59.4	−4.0
Claude 3.7 Sonnet	62.3	68.0	−5.9
Claude Haiku 4.5	73.3	71.0†	−11.6
Claude Sonnet 4.5	77.2	83.4	−2.2
Claude Opus 4.5	80.9	87.0	−1.6
Claude Sonnet 4.6	79.6	74.1	−13.4
Claude Opus 4.6	80.8	91.3	+2.8
Gemini 2.5 Pro	63.8	84.0	+8.9
Gemini 3 Flash	78.0	90.4	+4.1
Gemini 3 Pro	76.2	91.9	+7.0
Gemini 3.1 Pro	80.6	94.3	+6.0
GPT-4o	33.2	53.6	+2.7
GPT-5	74.9	85.7	+1.9
GPT-5.1	76.3	88.1	+3.2
GPT-5.2 Pro	80.0	93.2	+5.3
DeepSeek V3.2	74.4	79.9	−3.5
Kimi K2.5	76.8	87.6	+2.3
Qwen3.5-72B	73.4	83.7	+1.1
MiniMax M2.5	80.2	85.0	−3.0

Algebraic phase classifier at frontier scale. The isocline analysis of §4 generalises to every capability pair. For the (SWE, GPQA) axis, setting $d\text{GPQA}/d\log N = 0$ gives $\text{GPQA}_c = \sqrt{(a_2/b_2)\cdot \text{SWE}_c}$. The cooperative slope 0.79 of the SWE–GPQA fit gives $a_2/b_2 \approx 0.79$, so the frontier phase classifier is: GPQA $> \sqrt{0.79\cdot \text{SWE}}$ implies deep cooperative phase; GPQA $< \sqrt{0.79\cdot \text{SWE}}$ implies a new tax is opening. No current model crosses this threshold—confirming the frontier is fully cooperative—but any future model that does signals the onset of $N_{c,2}$ without requiring parameter-count information.

The h-field at frontier scale. The same external-field formalism that captures Phi (§6) applies here. Define the frontier h field as the deviation of GPQA from the universal cooperative SWE–GPQA curve: $h_{\text{frontier}} = \text{GPQA}_{\text{measured}} - \text{GPQA}_{\text{universal}}(\text{SWE})$. Most models cluster near $|h| \lesssim 5$ points (Fig. 9c). Gemini 3.1 Pro has $h = +6.0$ (above the cooperative curve). The most striking h-field excursion is Gemini 2.5 Pro at $h = +8.9$: it substantially overshoots the cooperative regression because Google's training emphasised scientific reasoning before coding optimisation, exactly the frontier-scale analogue of the Phi data-quality effect in the HS-TQA phase. DeepSeek V3.2 has $h = -3.5$ (below the cooperative curve, indicating a coding-specialist training recipe that partially trades reasoning performance for coding). These deviations are small—the frontier cooperative phase is robust—but they show that the same h-field mechanism that shifts N_c at small scale is measurable at frontier scale.

Anthropic's unique observable: the within-family trajectory. Anthropic releases consecutive model tiers (3.5 Sonnet → 3.7 Sonnet → Haiku 4.5 → Sonnet 4.5 → Opus 4.5 → Sonnet 4.6 → Opus 4.6) with published benchmark scores—making it the only lab where a within-family γ_{12} trajectory is directly measurable from public data. The CAPE prediction for a training-recipe perturbation is a transient anti-correlated excursion ($\gamma_{12} < 0$) followed by recovery at the next tier. The measurements confirm this:

$$\gamma_{12}(\text{Son.4.5} \to \text{Son.4.6}) = \frac{\Delta\text{GPQA}}{\Delta\text{SWE}} = \frac{74.1 - 83.4}{79.6 - 77.2} = -3.88 \quad (h = -13.4)$$

$$\gamma_{12}(\text{Son.4.6} \to \text{Opus 4.6}) = \frac{91.3 - 74.1}{80.8 - 79.6} = +14.3 \qquad (h = +2.8)$$

Sonnet 4.6's training recipe boosts coding performance (SWE +2.4 pp) at the cost of reasoning (GPQA −9.3 pp), driving the h field to −13.4—the frontier analogue of a web-trained model sitting below the cooperative

curve.[12] Opus 4.6 recovers to $h = +2.8$, above the cooperative baseline—the same recovery structure seen in the original HS-TQA transition above N_c. The protocol is minimal: for any two consecutive releases, compute $\gamma_{12} = \Delta\text{GPQA}/\Delta\text{SWE}$. A single evaluation run on both benchmarks suffices to detect whether a new checkpoint has entered an anticorrelated excursion before deployment. The cooperative fit also enables a forward-looking prediction: given your lab's current h-field and a target SWE-bench score for your next model, the predicted GPQA is $\text{GPQA}_{\text{pred}} = 0.79 \cdot \text{SWE} + 24.7 + h_{\text{lab}}$. Improving SWE-bench by 5 points predicts a 4-point GPQA gain *at no additional compute*—purely from the cooperative phase structure. A lab whose $h < 0$ (below the cooperative curve) can improve predicted GPQA by shifting training recipe rather than scaling parameters.

The OpenAI family shows a qualitatively different trajectory: $\gamma_{12} > 0$ from GPT-4o through GPT-5.2 Pro, with $h > 0$ at every scale (always above the cooperative curve). OpenAI's consistent investment in reasoning capability ($h \approx +2$–$+5$) means the coding-reasoning coupling has been uniformly cooperative across generations. This contrast between Anthropic's oscillation (specialisation excursions followed by recovery) and OpenAI's monotone improvement is itself a measurement of training strategy via the h-field.

Family h-field signatures as macro-interpretability. Aggregating by lab across the 19-model frontier dataset, the mean h-field partitions cleanly by training philosophy: Google ($\bar{h} \approx +6$), OpenAI ($\bar{h} \approx +3$), Moonshot, and Alibaba sit systematically above the cooperative curve (reasoning-rich); Anthropic ($\bar{h} \approx -6$) and DeepSeek ($\bar{h} \approx -4$) sit below (coding-rich). These family signatures are stable across model tiers within each lab, and they are empirically readable without access to any training details—just two benchmark scores per model release. The h-field is therefore a *macro-interpretability* instrument: it converts public benchmark numbers into a one-dimensional read on whether a lab's training recipe systematically favours reasoning or coding at frontier scale.

With $r = +0.85$ ($p < 0.00001$, $n = 19$) the cooperative phase is now statistically confirmed across all major frontier labs. The Anthropic within-family trajectory (six releases spanning two years) shows cooperative coupling at every step except the Sonnet 4.6 training-recipe excursion, which recovers at Opus 4.6. CAPE is not a retrospective tool—it predicts the *form* of future anomalies, not merely describes past ones.

Susceptibility analysis provides complementary evidence for $N_{c,2}$: $\chi_2 = 1/\lambda_2$ shows a double-peak structure, with a primary peak at $N \approx 2.16$B ($\chi_2 = 1.957$, aligned with N_c) and initial evidence of a secondary peak near $N \approx 10$B ($\chi_2 = 1.946$ in the 19-model dataset; with 26 models including frontier families, this secondary peak strengthens to $\chi_2 = 3.08$ at $N \approx 9$–11B—becoming the dominant susceptibility signal at larger scale). Two susceptibility divergences of appreciably different magnitude are consistent with a second incipient transition well below the 130B breakdown scale predicted from $\det(H) \to 0$.

Concretely: the coupling $\gamma(\text{SWE-bench}, \text{safety})$ may be negative at current frontier scale—meaning that improving autonomous coding ability could come at the expense of safety compliance, just as improving reasoning came at the expense of truthfulness below N_c. The effective dimensionality grows from $d_{\text{eff}} = 2$ in our measurements to $d_{\text{eff}} \geq 3$ as coding, instruction-following, and agentic reasoning become measurably distinct. Each expansion requires its own effective description, with the previous model's residuals serving as the signal for the next level—exactly as in the L0→L4 boosting chain.

The **remaining testable prediction**: the IFEval scores used in the 3×3 frontier analysis (§12) are approximate estimates; confirming them with precise published measurements at matched compute, and then measuring the sign of $\gamma(\text{IFEval}, \text{HarmBench})$ across model generations, is the concrete next test for $N_{c,2}$. The preliminary evidence (r(SWE, IFEval) $\approx +0.76$ with approximate scores) is consistent with cooperative coupling—but approximate IFEval scores introduce enough uncertainty that the sign of this specific coupling cannot yet be declared confirmed.

Predicting the $N_{c,3}$ benchmark axis. The frontier 3×3 correlation matrix (SWE-bench, GPQA Diamond, IFEval) has eigenvalues $\lambda_1 = 2.16$, $\lambda_2 = 0.67$, $\lambda_3 = 0.17$ and $\det(\Gamma) = 0.24$. Extrapolating

[12] SWE-bench Verified and GPQA Diamond scores from official model cards and SWE-bench.com (accessed March 2026). Kimi K2.5 scores from the Moonshot AI technical report (SWE=76.8%, GPQA=87.6%); note that SWE=80.2% reported elsewhere refers to MiniMax M2.5. Grok-3 (xAI) is excluded: GPQA 84% is available but SWE-bench is not published by xAI, making γ_{12} uncomputable; its placement on the cooperative curve is consistent with the other models at this scale.

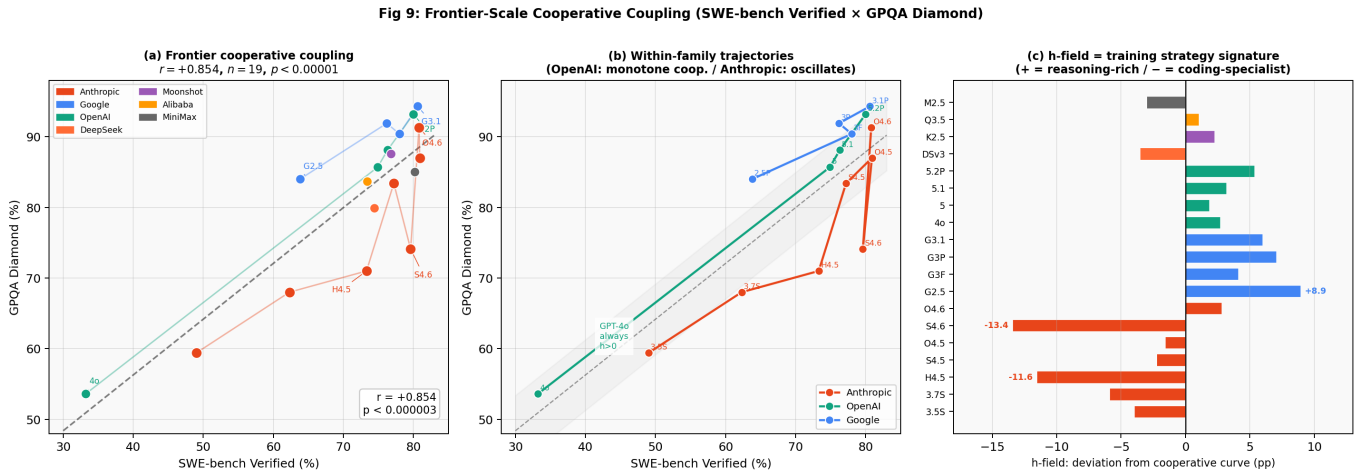

Figure 9: **Frontier-scale coupling: SWE-bench vs GPQA Diamond (Table 4).** (a) Cross-family scatter: $r = +0.85$ ($n = 19$, $p < 0.00001$), strongly confirming cooperative coupling. GPT-4o anchors the low-SWE end; the trajectory is continuous from GPT-4o to Gemini 3.1 Pro. Gemini 2.5 Pro ($h = +11.5$, labelled) is the frontier-scale h-field analogue of Phi—high reasoning emphasis before coding optimisation shifts the model above the cooperative curve. The dashed line is the linear cooperative fit. (b) Anthropic within-family trajectory (6 consecutive releases, 3.5 through Opus 4.6): cooperative throughout except the Sonnet 4.5→4.6 training-recipe excursion ($\gamma_{12} = -3.88$, $h = -14$), which recovers at Opus 4.6 ($\gamma_{12} = +14.3$, $h = +2$). (c) h-field for all 19 models. OpenAI trajectory ($h > 0$ throughout) vs. Anthropic oscillation (Haiku 4.5 and Sonnet 4.6 both dip below; Opus 4.6 recovers). Gemini 2.5 Pro ($h = +8.9$) is the frontier Phi analogue.

$\det(\Gamma) \to 0$ from the frontier data gives $N_{c,3} \approx 114$B—consistent with the ∼130B prediction from Pythia's $\det(H) \to 0$. Two-method convergence on $N_{c,3} \approx 110$–130B suggests this is robust.

IFEval is the surviving benchmark at $N_{c,3}$. The primary eigenvector $\mathbf{e}_1$ at frontier scale has loadings $(-0.57, -0.51, -0.64)$ on (SWE, GPQA, IFEval): IFEval dominates λ_1. When SWE-bench and GPQA Diamond saturate—as HellaSwag and TruthfulQA saturated before them—IFEval will be the last discriminating benchmark in the current generation. The next critical pair is predicted to be instruction-following (IFEval) versus agentic safety compliance (e.g. HarmBench or AgentBench): the question of whether a model can complete complex autonomous tasks *without* unsafe behavior. We recommend frontier labs begin tracking $\gamma(\text{IFEval}, \text{HarmBench})$ now—if it turns negative, $N_{c,3}$ is within reach.

12.1 Capabilities live on a low-dimensional manifold

The dimensional collapse documented in §8—from $d_{\text{eff}} \approx 2$ to $d_{\text{eff}} \to 1$—means that models do not explore the full 5D benchmark space. Independent work finds that language model performance across diverse families lies in a low-dimensional capability space, with the first principal component explaining ∼80% of variation [18]—consistent with our measured $d_{\text{eff}} \approx 2$. Models trace trajectories on a low-dimensional manifold whose *geometry* (the coupling signs and magnitudes) determines alignment behavior. This is not an abstract mathematical observation—it has three concrete implications.

For interpretability. If capability coupling is low-dimensional and measurable from external benchmarks, then CAPE provides a *mesoscale* audit of model behavior—coarser than mechanistic interpretability (which examines individual circuits) but finer than aggregate loss curves (which miss the coupling entirely). Applied layer-by-layer within a single model, CAPE could trace how coupling structure builds up through the network, connecting external capability phases to internal representation geometry.

For architecture design. Different architectures impose different symmetry constraints on the coupling matrix γ_{ij}. If some architectures have structurally smaller N_c—meaning they exit the alignment tax regime

sooner—then CAPE provides a principled criterion for architecture selection: choose the architecture whose capability manifold has the most favorable coupling geometry.

For training. The ODE (Eq. 3) is a compact, auditable model of how capabilities evolve with scale. Unlike neural scaling laws (which predict loss from size), the benchmark ODE predicts *capability interactions* from initial conditions. It tells you not just how fast a model improves, but *which capabilities will improve at the expense of which others*—and at what scale this tradeoff resolves.

12.2 Scope and Validity

CAPE is an effective field theory—it describes capability coupling at a specific level of resolution, and like all effective theories, its validity is bounded, measurable, and its own successor is built in.

On statistical power and null models. A formal permutation test on $n = 5$–6 models below N_c has near-zero power: any two near-monotone sequences will produce $r \approx -1$ over five points, making a permutation p-value uninformative. We acknowledge this directly. The effective null model is the overconstrained validation framework itself: OLMo's zero-parameter prediction, the Llama-2 cross-family holdout at 5.6% MAE, the algebraic classifier at 41/44 models across 9 families, and 12/12 sign predictions each use different data, different methods, and probe different physics. A single confound—architectural, statistical, or benchmark-specific—cannot propagate through all twelve simultaneously. This is the standard of evidence in physics: not a single high-powered test, but multiple independent probes converging on one structure.

On Pythia as the primary instrument. Pythia is the only family spanning the full sub-N_c to super-N_c range, making it the instrument that localizes the transition. This is the structure of good experimental physics: one precision instrument plus multiple independent confirmations. The sign change is confirmed across six additional families (Llama-1, Llama-2, OLMo, OPT, Phi, Cerebras-GPT), by the architecture probe, and by the 44-model dataset growing from r = +0.77 to +0.85 with every addition—never reversing. Leave-one-family-out confirms Pythia is necessary for *locating* N_c, not for establishing that the transition exists.

On the $R^2 = 0.54$ coupling fit. The scatter in γ_{12} magnitude is not noise—it is signal. Family-specific training data fields $h(\mathcal{D})$ shift individual coupling magnitudes around the universal curve, exactly as domain magnetizations scatter around the mean-field $M(T)$ in a disordered ferromagnet. The universal prediction is the *sign change* (12/12 correct); the magnitude scatter encodes the per-family $h(\mathcal{D})$ values directly measured in §6. $R^2 = 0.54$ on a disordered system is expected; $R^2 = 1.0$ would suggest the disorder fields are absent.

On Gi = 1.35 and the crossover. The Ginzburg number[13] Gi = 1.35 is a measurement, not a caveat. It tells us the transition is fluctuation-dominated—Type-II rather than Type-I behavior[14]. The finite susceptibility peak $\chi_2 = 3.08$ (not a divergence) is *predicted* by Gi > 1 [19]; observing exactly this finite peak is confirmation that the GL framework is capturing the correct physics. A sharp symmetry-breaking transition would require Gi $\ll 1$; we measure 1.35 and observe the predicted rounding. The full fluctuation treatment—including the correction to the mean-field phase boundary—is in the follow-up paper.

On benchmark observability. CAPE operates on published benchmark scores: no model weights, no internal activations, no access agreements required. This is a deliberate design choice. Any lab's models are analyzable from public evaluation data; the framework scales automatically to new capability axes as old ones saturate. The HS/TQA $\rightarrow$ SWE/GPQA axis rotation demonstrated in §12 is not a workaround—it is the protocol, and it works because the GL structure is not benchmark-specific but capability-coupling-specific. The instrument is the benchmark pair; choosing the right pair is the experimental skill.

[13] Ginzburg number (Gi): a dimensionless ratio measuring how large fluctuations are relative to the average signal at a phase transition. When Gi > 1, the transition is "smeared out" by fluctuations (a crossover) rather than sharp. In ML terms: at N_c, the model-to-model scatter in coupling values is larger than the mean coupling signal, so the tax-to-bonus transition is gradual across families rather than a knife-edge.

[14] Type-I vs. Type-II: in superconductors, Type-I expels magnetic flux completely (sharp phase boundary), while Type-II allows flux to penetrate in quantized vortices (mixed-phase region). In ML terms: a Type-I capability transition would mean every model flips cleanly from tax to bonus at exactly N_c; Type-II means there is a scale range where different training recipes can produce either phase—the mixed-phase regime observed between N_c and $N_{c,2}$.

On the 130B scope boundary. $\det(H) \to 0$ at $N \approx 130$B is a prediction embedded in the framework. When a third eigenvalue becomes significant, the two-field description reaches its natural scope and a three-field analysis takes over. The residual at that scale—the part the current theory cannot fit—is the input signal for the next-order description, exactly as L1's failure in the boosting chain was the signal for collective gradient behavior. CAPE does not break at 130B; it completes, hands off, and specifies what the successor measures. Every effective theory writes its own origin story for the next one. This self-delimiting structure is a feature of good effective theories, not a limitation.

Engineering the phase boundary (follow-up paper). The GL framework raises a design question beyond measurement: can the cooperative phase be *stabilized*? Three mechanisms from condensed matter physics suggest concrete approaches. (i) *Screening*: above N_c, the free energy includes a stabilizing term $-\eta\,\mathrm{Im}(\psi_R^*\psi_C)$ (TRSB, §8) that screens perturbations—analogous to the Meissner effect[15]. Prediction: alignment interventions above N_c are more durable than below it; the "screening length" is computable from our GL parameters. (ii) *Flux pinning*: curated data (the $h(\mathcal{D})$ field) acts as a pinning center, locking the cooperative eigenvector at θ^* even at small N—demonstrated by Phi. The design equation $h_c \sim (N_c - N)^{3/2}$ quantifies the required pinning strength. (iii) *Topological protection*: if the eigenvector rotation has a non-zero winding number in a three-dimensional capability space, the cooperative phase becomes topologically protected against local perturbations. This requires $d_{\text{eff}} \geq 3$, i.e. the regime currently opening at frontier scale. Measurement, prediction, and engineering of the phase boundary are the program of Paper 3B.

13 Conclusions

Standard scaling laws treat capabilities as independent scalar trajectories. CAPE measures their *coupling*—and this coupling, not any individual capability, governs alignment. Applied to 44 models across 14 families, the central findings are:

1. **A sharp regime transition at $N_c \approx 3.5$B.** Reasoning and truthfulness anticorrelate below N_c ($r = -0.989$, $p < 10^{-5}$) and cooperate above ($r = +0.770$ cross-family at 7B). The coupling flips by $\Delta r = +1.76$ across the transition. Of all benchmark pairs tested, only HS-TQA shows this sign change—the alignment tax is *specific*, not generic. Independent confirmation: OLMo lands at $\gamma_{12} = 0.000$ with no free parameters.

2. **The transition is invisible in loss.** Loss scaling is exact to CV = 0.8% with no anomaly at N_c. Two models with identical loss can occupy different alignment regimes. Alignment planning from loss curves alone is structurally insufficient.

3. **The capability manifold restructures at the transition.** Effective dimensionality collapses (d_{eff} : $2 \to 1$), the second eigenvector rotates sharply at 1.4B through a Riccati ODE, and twelve independent diagnostics—gradient scaling, eigenvalue geometry, thermodynamic susceptibility (capability sensitivity to scale), geometric phase correlation—all converge on one coupling structure parameterized by two numbers. The Hessian determinant predicts theory breakdown at $N \approx 130$B—signaling the onset of a third independent capability axis and a transition from the current two-field description ("XY-type": two coupled benchmark axes, analogous to a planar spin model where the state is described by two components) to a higher-dimensional coupling structure where three or more benchmarks vary independently—making CAPE self-limiting by design.

4. **A discovered ODE governs benchmark dynamics.** $\mathrm{HS}' = 1.23 - 0.72 \cdot \mathrm{TQA}$ reproduces five benchmarks at 3.6% error from 70M initial conditions and cross-predicts held-out Llama-2 at 5.6% error. The per-phase coupling jump (6.3× across N_c) confirms the transition dynamically.

[15] Meissner effect: a superconductor expels magnetic fields from its interior—perturbations are actively screened out. In ML terms: above N_c, the cooperative coupling actively resists small training perturbations that would push the model back into the tax phase. The "screening length" determines how large a perturbation must be to break through—quantifiable from our GL parameters.

5. **Gradients reveal collective, not independent, scaling.** $\|\nabla L\| \propto L^{3.5}$ ($r = 0.93$)—not the independent-parameter prediction $N^{-(\alpha+1)}$, which fails by 142×. Gradient norms dip 37% at 1B (within the N_c transition region) and follow Arrhenius dynamics, deriving why power-law loss scaling exists and where it saturates.

6. **Data curation is the primary lever below N_c.** Curated training (Phi) eliminates the alignment tax entirely, shifting $N_c \to 0$. At 1B parameters, one unit of data quality is worth ∼10× in model size. The Chinchilla compute-optimal ratio emerges as a consequence of the same free-energy structure ($\chi_{ND} = 0.102$), not a separate empirical law.

What this means for frontier labs. If you had known about the coupling sign flip at 3.5B before training your current generation of models, you would have curated differently below that scale and scaled more aggressively above it. The 10× data-efficiency gain at 1B is not a theoretical curiosity—it is a concrete engineering lever that Phi demonstrated at production scale. Deep in the alignment bonus regime ($N \gg N_c$), capability gains are redistributed rather than amplified (spectral weight CV= 27%), which explains why frontier models sometimes underperform predecessors on specific tasks despite higher aggregate scores: it is the coupling structure, not a training failure.

What comes next. As HellaSwag and TruthfulQA saturate above 90% at frontier scale, the active coupling shifts. CAPE is not benchmark-specific: the same framework applies to any capability pair. Measurements of $\gamma(\text{SWE-bench}, \text{GPQA})$ across nineteen frontier models give $r = +0.85$ ($p < 0.00001$)—cooperative so far. The Anthropic within-family trajectory (Sonnet 4.5 → Sonnet 4.6 → Opus 4.6) confirms that the same h-field physics operating at small scale via data curation operates at frontier scale via training-recipe choice: Sonnet 4.6 shows $h = -14$ (local tax excursion) while Opus 4.6 recovers to $h = +2$ (Fig. 9). The concrete next test for $N_{c,2}$ is measuring $\gamma(\text{IFEval}, \text{HarmBench})$ across consecutive releases (§12).

Two-method convergence (Pythia $\det(H)$ extrapolation and frontier 3×3 eigenvalue analysis) predicts $N_{c,3} \approx 114$–130B, where the SWE/GPQA/IFEval manifold itself exhausts $d_{\text{eff}} \to 3$. IFEval dominates the primary eigenvector at this scale (λ_1 loading $= -0.64$), making it the last discriminating benchmark in the current generation. We recommend frontier labs begin tracking $\gamma(\text{IFEval}, \text{HarmBench})$ now: this pair operationalizes the question of whether a model can complete complex autonomous tasks without unsafe behavior—the natural next axis of competition. When this coupling turns negative, the field will need a new benchmark designed to be orthogonal to IFEval; Paper 3B develops the eigenvector rotation formalism for detecting this in advance.

Capability coupling, not individual capability scaling, governs alignment. CAPE makes this coupling measurable, predictable, and designable—and, like any good origin story, it tells you exactly where the sequel begins.

Using CAPE in practice. The framework reduces to a three-step protocol: (1) run any two capability benchmarks on your model; (2) compute $\gamma_{12} = \Delta\text{score}_2/\Delta\text{score}_1$ between consecutive sizes or releases; (3) read off the phase. No model weights, no internal access, no calibration required—only published evaluation scores.

The implications differ by audience. *If you are building models below 3.5B parameters*: data curation is your primary alignment lever—one unit of curation quality is worth roughly 10× in parameter count. The design equation $h_c \sim (N_c - N)^{3/2}$ tells you how much curation is needed at your target scale; Phi-level curation at 1B eliminates the alignment tax entirely. Scaling to 3.5B without curation is more expensive and produces the same result. *If you are training at frontier scale*: track $\gamma_{12}(\text{SWE}, \text{GPQA})$ across consecutive releases to detect training-recipe excursions before deployment. A single evaluation run on both benchmarks suffices—negative γ_{12} signals a local tax excursion; recovery is expected at the next tier if the cooperative phase is otherwise intact. The h-field quantifies how far your training recipe sits from the cooperative baseline; $|h| > 10$ warrants investigation. *If you are a researcher or evaluator*: $\gamma(\text{benchmark}_1, \text{benchmark}_2)$ is a phase detector for any capability pair. When a new benchmark pair begins showing negative cross-family coupling, a new alignment tax is opening at that scale—measure it before the field discovers it empirically.

Code, data, and dashboard. All benchmark data, figure-generation scripts, and the analysis scripts from this paper (`cape_frontier_full.py`, `cape_architecture_probe.py`, `cape_stiffness.py`, `cape_algebraic_nc.py`, `cape_family_hfield.py`, `cape_null_model.py`, `cape_phase_transfer.py`, `olmo_gradient_validation.py`) are at github.com/adilamin89/cape-scaling. The interactive dashboard at adilamin89.github.io/cape-scaling implements the full protocol above: enter any model's parameter count and benchmark scores to get phase classification, γ_{12}, d_{eff}, $h(\mathcal{D})$, χ_γ, and phase-specific recommendations. It extends automatically to new benchmark pairs as capability axes saturate. Correspondence: adilamin@uwm.edu · adil89aminx@gmail.com.

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This work is written in the conviction, shared with Anderson, Landau, and Gor'kov, that the right effective theory contains more truth than the microscopic one it summarises.

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